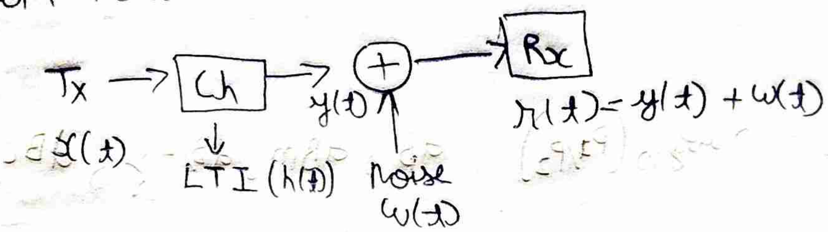
Signals recap ~~of~~ Lecture 2

- Why do we need complex valued signals? When signals are real? → are transported
  ~~because analysis~~ ~~ter~~ because analysis ~~tof~~ becomes easier.
  $X(t) \rightarrow X(\omega)$ → has a complex representation
  ↳ complex

- Sampling of continuous signals is lossless if signal is bandlimited.

- After sampling, you do quantization (Acts as a channel) which introduces random noise
  ↳ additive noise

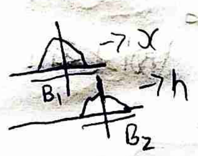


- noise is gaussian due to central limit theorem.

- We assume channel is LTI, so we can characterize it using $h(t)$.
  Why is it this assumption valid? (It is an approximation)

- If $y(t) = x(t) * h(t)$ → what is the bandwidth of $y(t)$
  $B_1 \quad B_2$

  $y(t) \leq \min(B_1, B_2)$



- We assume we have Energy signals and not Power signals in this course
  → Page 61 of textbook

- f = frequency, $|X(f)|^2$ → energy spectral density
  ↳ used to reduce noise in or something on oscilloscope
  ↳ check again

  in a band $\int_{f_1}^{f_2} |X(f)|^2$

  access
- TDMA (time division Multiplexing), TDD
- FDMA (freq division Multiplexing)
- SDMA (Space) → antenna directivity → ~~cell~~ cellular systems

- we want to reduce energy spectral density in other bandwidths
  ↳ for example in FDM it might ~~interfear~~ interfear with other bandwidth channels where other people may be communicating

- In all of these Multiplexing we want to ensure the signals transmitted are orthogonal.

- If something is orthogonal in time is it also in freq → check
  ↳ $\int x_1(t)\, x_2^*(t)\, dt = 0$

• CDMA → Code division Multiplexing → phase difference (sin, cos)

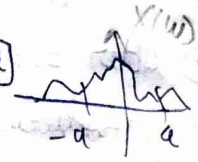
Band limited Signals → Uncertainity Principle

• we can never get Band limited signals → means ∞ time domain $x(t)$

• we can only generate time limited signal → ∞ ~~spectro~~ spectrum component

Definitions of bandwidths ↓

• 90% band width → 90% of energy spectral density $[-a, a]$



• Decibel : $10 \log_{10}(P_1/P_2)$, dB , dBm , dB/Hz , dBC

dBm → mW

dBC → carrier freq

• 3dB bandwidth → where our power becomes half

• ~~Rayl~~ Paley - Wiener Criterion , → to check if a filter is realizable

$$\int_{-\infty}^{\infty} \frac{\ln|H(\omega)|}{1+\omega^2} < \infty$$

~~New~~ Lecture 3

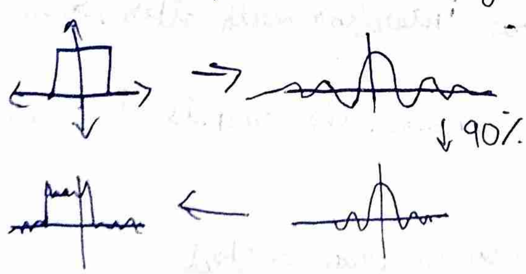
• Time varying Convolution → for LTV systems.

• We model communication systems as $r(t) = h(t) * m(t) + n(t)$

$r(t)$ → reciever, $h(t)$ → Channel, $m(t)$ → message, $n(t)$ → AWGN

• We use orthogonality to seperate carrier signals, as by taking their projection $\int (m_1(t) + m_2(t))\, m_2^*(t)\, dt$

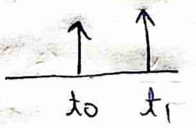
• In real life we can only generate Linear phase ~~systems~~ filters



- All types of filter that we ~~use~~ use we put them into the channel.
  at the reciever or transmitter

- A simple wired channel can be modeled as
  $h(t) = a_0 \delta(t - t_0)$ → attenuation and delay due to distance

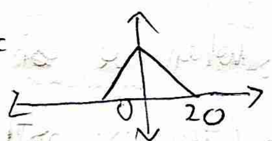
- but sometime due to reflections from objects in a wired communication we get $h(t) = a_0 \delta(t-t_0) + a_1 \delta(t - t_1) \ldots$

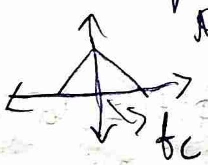
we get let say  → in time we get maxima and minima

- This is called Inter Symbol Interfearence (ISI)
- We can construct Ideal filters in digital domain → Why? because we made an assumption that $x[n]$ is periodic, (but we look at one period) so we see ~~sharp~~ sharp spectrum.
  → Check

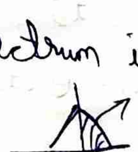
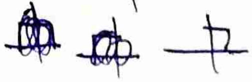
Lecture 4

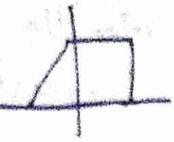
Recap
→ we use GHz freq because of antenna length constraints.

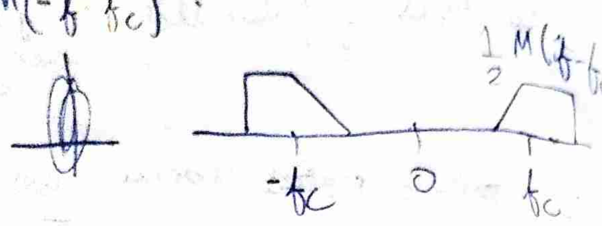
- $M(\omega) =$  → we want to convert this to a higher freq for transmission → Up conversion

-  $= M(f - f_c)$ → carrier frequency

- The equivalent representation is $m(t)e^{j2\pi f_c t}$

- $m(t)$ can be complex as well, $m(t) = m_{re}(t) + j m_{img}(t)$
  advantage, for a real signal the spectrum is symmetric so the info on one side is redundant  redundant so you can combine signal, $m_1 =$  $m_2 =$ 
  $\therefore m_1 + j m_2 =$ 

• $M(f) =$    Scale $\frac{1}{2}M(-f-f_c)$:

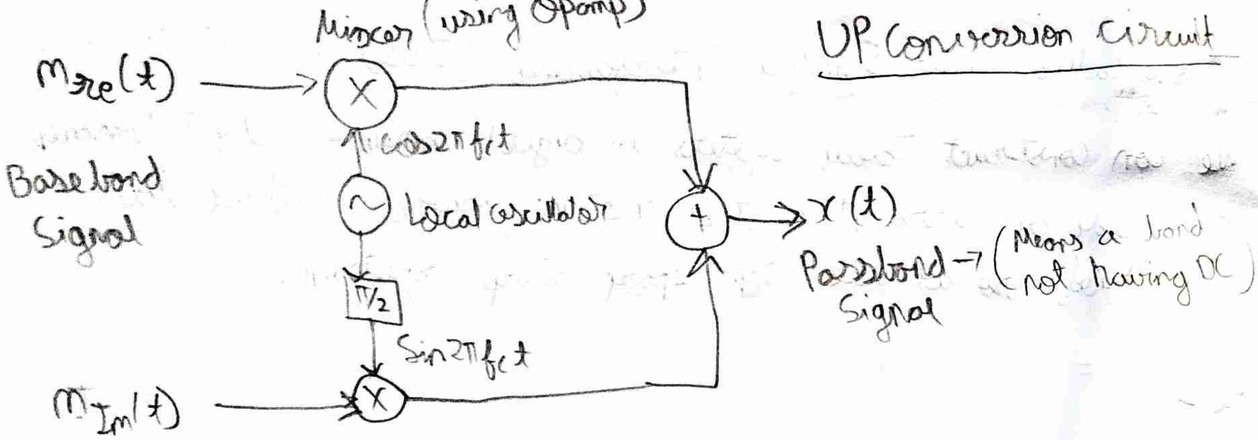


$$\underset{\uparrow}{x(t)}$$
for $Re\{m(t)e^{j2\pi f_c t}\} = \frac{z(t)+z^*(t)}{2}$, $z(t) = m(t)e^{j2\pi f_c t}$

$M^*(t) \longrightarrow m_{re}(t)\cos 2\pi f_c t + m_{Im}(t)\sin 2\pi f_c t$

$x(t) = m_{re}(t)\cos 2\pi f_c t - m_{Im}(t)\sin(2\pi f_c t)$

UP Conversion circuit



• $\cos 2\pi f_c t \perp \sin(2\pi f_c t)$ (orthogonal) → ask again, when we say modulating we are modulating amplitude of both of them (they are orthogonal)

• The phase is also being modulated

• We define an envelope $e(t) = \sqrt{m_{re}^2(t) + m_{Im}^2(t)}$, $\phi(t) = \tan^{-1}\left(\frac{m_{Im}(t)}{m_{Re}(t)}\right)$

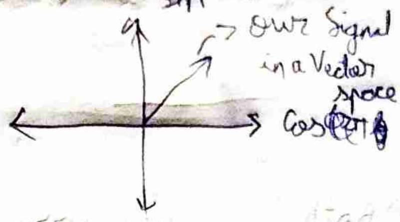
$\therefore m_{re} = e(t)\cos\phi(t)$, $m_{Im} = e(t)\sin\phi(t)$

Substitute in $x(t)$

to get $x(t) = e(t)\cos(2\pi f_c t + \phi(t))$

• m_{Re} and m_{Im} can be arbitrary in time domain

• In freq, $f_c >> B$ because high bandwidth will spread to a freq which our antenna will not transmit, so we have this restriction.

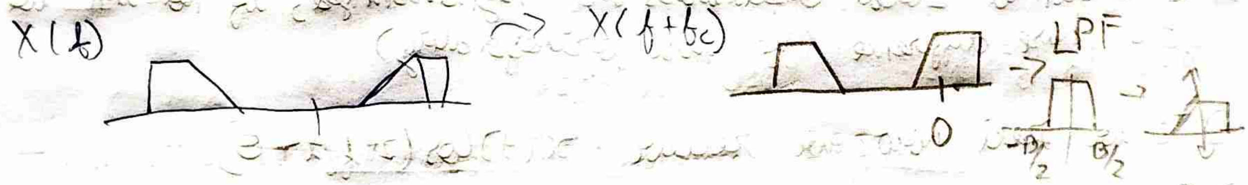


• $m(t) = m_I(t) + j\, m_Q(t)$

m_I ↓ Inphase

m_Q ↓ Quadrature Phase

• Now we have to ~~transmit~~ invert this signal



• But IRL we cannot multiply with imaginary freq

we do $Re(2x(t)e^{j2\pi f_c t})$

2 ↓ Scaling

$\Rightarrow 2x(t)\cos 2\pi f_c t = (1+\cos 4\pi f_c t)\, m_I(t) - m_Q(t)\sin 2\pi f_c t$ — $\sin 4\pi f_c t$?

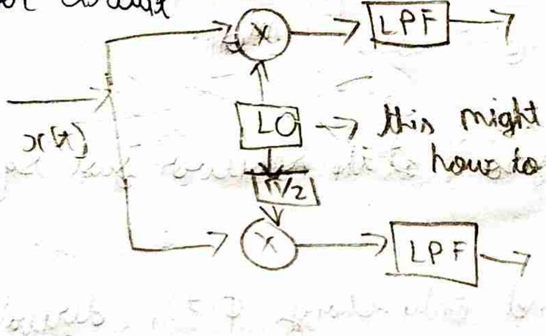
LPF $\rightarrow m_I(t)$

to get other component, ~~$-2x(t)\cos($~~ ~~$-2\cos$~~

$-2x(t)\sin 2\pi f_c t =$ ~~$-2m_I(t)\sin(2\pi f_c t)\cos(2\pi f_c t)$~~

Receiver Circuit

Down Conversion Circuit



→ this might be out of Phase with the transmitter
have to deal with it?

→ Because we can have finite duration signals, our spectrum will be ∞, so we ensure 99% of energy in the region of interest. But if we have multiple antennas talking then 1% from each may add up.

Lecture 5

- If we use wired medium we are sending only 1 signal, are we sending less information ~~compa~~ compared to wireless.?
  No as in wireless we use more bandwidth to send 2 signals

- Both the upconversion and downconversion blocks are exactly the same but how are we able to recover the signal. → ?
  ↳ Think about this

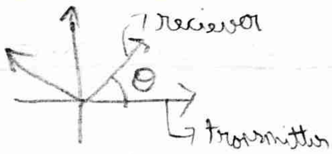
- We assumed Local oscillators are synchronized, if not then we get a phase difference (we loose orthogonality)

- ~~If we reci~~ now we recieve $x(t)\cos(2\pi f_c t+\theta)$
- transmit power amplifier :- multiply $x(t)$ at the transmitter by 2 to avoid scaling at the sender (noise is amplified)

$$m_I(t)\left[\cos(4\pi f_c t+\theta)+\cos\theta\right]$$
$$-m_Q(t)\left[\sin(4\pi f_c t+\theta)+\sin\theta\right]$$

after LPF, $I \to m_I(t)\cos\theta - m_Q(t)\sin\theta$
$Q \to m_I(t)\sin\theta + m_Q(t)\cos\theta$

we recieve a rotated axis at the reciever



$$\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}m_I\\ m_Q\end{bmatrix}$$

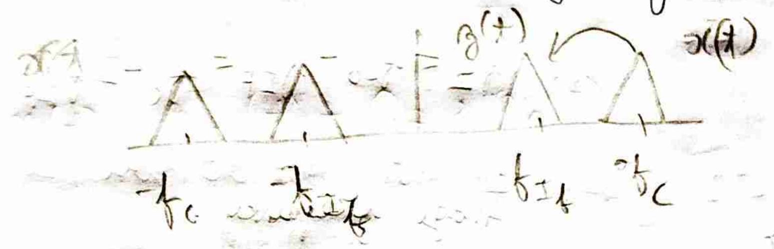
- Amplitude modulation, set M_Q to zero, at the reciever just square and add to get $m_I(t)$

- Phase modulation, set $e(t)$ to 1 and only change $\phi(t)$, derivative of $\phi(t)$ is the freq (freq modulation)

↳ we can use ~~phased~~ phase locked loops to synchronize

Hetrodyne & Super hetrodyne

- Our phones have too many bands of freq that they use. We get the carrier freq to an intermediate freq which is what our circuits are designed for.



$$z(t) = Re\left\{x(t)\, e^{j2\pi(f_c - f_{If})t}\right\}$$

do

- There was an issue with circuit that freq at $f_c - 2f_{IF}$ was also shifted to the $-f_{If}$ freq, put a BPF. In old AM radio stations they could hear 2 stations simultainiously.
- At the end you put a BPF at f_{If}

Lecture 6

- If LO are synchronized then you call it homodyne
- Superhetrodyne → Bringing down to intermediate frequency and then bring that down again to DC.
- I_{fc} is around 450 MHz
- at the reciever side we have $x(t)\cos(2\pi f_{LO} t)$

$$m_I(t) = \frac{m(t)}{2}\left[\cos(f_{LO}+f_c)t + \cos 2\pi(f_c - f_{LO})t\right]$$

$$m_Q(t) = -\frac{m(t)}{2}\left[\sin 2\pi(f_c + f_{LO})t + \sin(2\pi(f_c - f_{LO})t)\right]$$

we get $f_c - f_{LO} = f_{IF}$ → we convert to f_{If}

• We saw that this freq of ~~f~~ f_{LO} we had issues

$f_2 - f_{LO} = -f_{IF}$

$f_2 = f_{LO} - f_{IF} = f_c - 2f_{IF}$

↓

Image frequency.

↓

this is for $Re\{x(t)e^{-j2\pi f_{LO}}\}$

• here we are shifting to the ~~right~~ left, but we can also shift to the right.

↳ $f_c + f_{LO} = f_{IF} \Rightarrow f_{LO} = f_c + f_{IF}$

we want  we get 

• Why is it getting flipped? ⇝ see complex representation

$\boxed{\text{Spectral inversion}}$

$f_{LO} = f_c - f_{IF}$

$\frac{m_I(t)}{2}\cos$

$- m_Q(t)\sin$

$= \frac{1}{2}Re\{m(t)e^{j2\pi f_c t}\}$

$f_c + f_{IF}$

$\frac{m_I(t)}{2}\cos$

$+\frac{m_Q(t)}{2}$ ~~cos~~ $\sin$ → Check

$= \frac{1}{2}Re\{m^*(t)e^{j2\pi f_c t}\}$

• How to select from the above two

we had image frequency as $f_c - 2f_{If}$ before and now shifting right we have $f_c + 2f_{If}$

↳ so we can use a low pass filter

it is an Image ~~Reflection~~ Rejection Filter

• lets say $f_c \in [f_l, f_u]$ → Same antenna, different f_c operation

picking LO, 1

$f_{LO} = [f_l - f_{IF}, f_u - f_{IF}]$

$= [f_l + f_{IF}, f_u + f_{IF}]$

→ $\frac{f_l - f_{IF}}{f_u - f_{IF}}$

↓

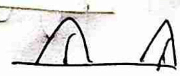
the ratio will be different

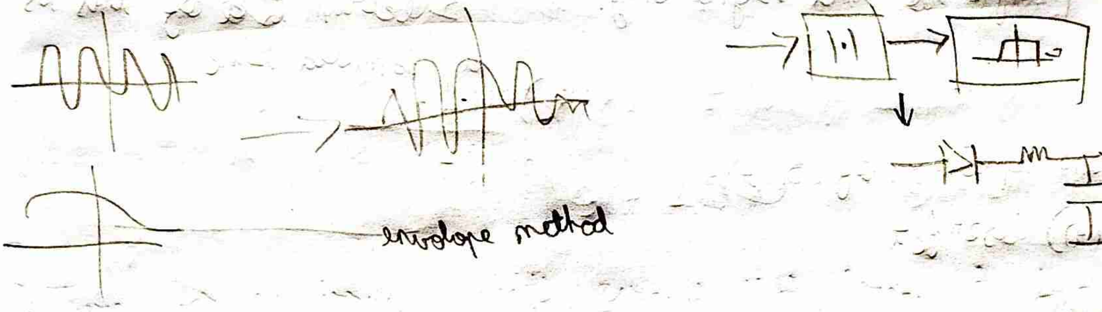
- That ratio is the swing of the filter and we want it to be small, small, even if the range of freq is the same.

Amplitude Modulation

- $x(t) = e(t)\cos(2\pi f_c t + \phi(t))$
- Let's say we don't want to worry about θ (Phase offset between filter). We can just set $m_Q(t) = 0$

then $x(t) = m_I(t)\cos(2\pi f_c t)$

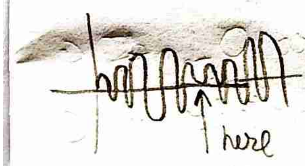
- AM is very inefficient, what we transmit  we are sending 4 times of what we need



- Issues with AM $\Rightarrow$ $m_I(t)$ cannot be -ve as |.| ignores it.

Lecture 7

Phase reversal issue, for -ve $m(t)$ there is a signal gets a discontinuity



DSB-SC

Double side band - Suppressed Carrier

- we can add an offset to $m(t)$
  - $\hookrightarrow$ $(m(t) + A_c)\cos 2\pi f_c$, with $m(t) + A_c \geq 0$ always

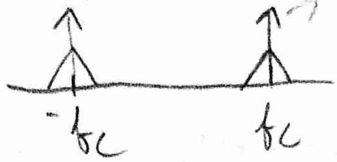
Amplitude of the carrier freq (DSB-AM) ↗ Amplitude Modulation

- If $\min m(t) = -M$, then modulation index $\alpha = \dfrac{M}{A_c} \leq 1$
  - $\hookrightarrow$ Also assume $M(t)$ has no DC components

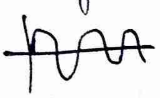
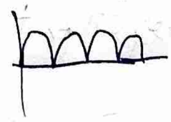
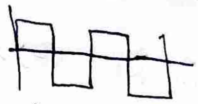
$\therefore$ $A_c(1 + \alpha\, m(t))\cos(2\pi f_c t)$

- DSB-SC spectrum

- DSB-AM spectrum

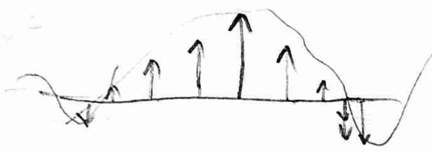


- the modulus is not an LTI system, but how do you see it mathematically

$x(t) \rightarrow$  , $|x(t)| =$  equivalent to multiplying with 

which is specific to this signal only.

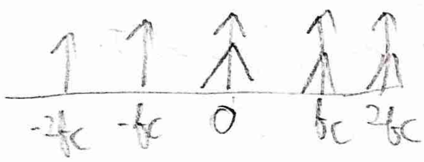
Spectrum of this is a sampled sine



so what is the spectrum of $x(t)$

$A_c(1+\alpha m(t)) \cos 2\pi f_c t$

$\downarrow$



- How inefficient is our AM? How to quantify it?

  ↳ $\dfrac{\text{Power of message}}{\text{Power of Tx signal}}$

$$x^2(t) = A_c^2 (1+\alpha m(t))^2 \cos^2(2\pi f_c t) \rightarrow \underset{\uparrow\, A_c \alpha m(t)}{\text{message}} + \underset{\uparrow\, A_c}{\text{Carrier}}$$

$$= \underbrace{\frac{A_c^2}{2} (1+\alpha m(t))^2}_{y(t)} (1+\cos(4\pi f_c t))$$

$y(t) \cos 4\pi f_c t$ after integrating is zero.

- Also our assumption that the bandwidth of message is less than f_c so we will definitely not get a DC component $\rightarrow$ Integral is zero

- the power of $x(t)$ is then $\frac{A_c^2}{2}(1+\alpha^2 P_m)$ — this term goes to zero why?

P_m → Power of $m(t)$

- Is ~~energy~~ Power in Passband the same as in the power in the baseband?
  → part is but it doesn't make AM or SC any efficient

- $x_m^2(t) = (A_c \alpha m(t) \cos 2\pi f_c t)^2$

this is in the Passband $= \frac{A_c^2}{2}\alpha^2 m^2(t)$ — has to be normalized

$$\eta = \frac{\text{Power of msg}}{\text{Power of } T_x \text{ signal}} = \frac{\frac{A_c^2}{2}\alpha^2 P_m}{\frac{A_c^2}{2}(1+\alpha^2 P_m)} \approx \frac{a}{1+a}$$

→ can never be 100%

η → Power efficiency

$$PAPR = \frac{\text{Peak of } m(t)}{\text{Avg Power of } m(t)}$$

→ it is energy and not Power (Misnomer)

PAPR → Peak to average Power Ratio

- maximum efficiency is 50%, but for this we assume $P_m = 1$ and $\alpha = 1$, but if you send a sinusoid, $P_m = 1/2$ and then you get even lower efficiency. this just ~~mea~~ shows there is a relation to PAPR

Lecture 8

- we had assumed, $\int m(t) = 0$, no DC component.
- How does the PAPR affect the whole of above setup.

$$PARP = \frac{\frac{A_c^2}{2}(1+1)}{\frac{A_c^2}{2}(1+\alpha^2 P_m)} \rightarrow \frac{2}{1+\alpha^2 P_m} = b$$

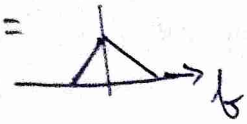
High PAPR is always bad

Power in $A_c(\alpha m(t)+1)\cos 2\pi f_c t$

now sub this in $\eta = 1 - \frac{b}{2}$, ↑PAPR, ↓η

• SSB (Single side band)

removing redundancy.

• $M(f) =$ 

$$M(f) = M(f)u(f) + M(f)u(-f)$$
$$M_+(f) + M_-(f)$$
$$\downarrow \qquad \downarrow$$
$$m(t) = m_+(t) + m_-(t)$$ because fourier transform is linear

• multiplying by $u(f)$ is convolution in time
↳ find out

• $X(f) = M(f-f_c)u(f-f_c) + M(f+f_c)u(f+f_c)$

$\therefore x(t) = m_+(t)e^{j2\pi f_c t} + m_-(t)e^{-j2\pi f_c t}$

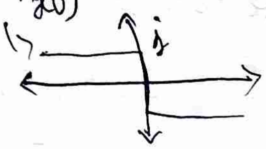
• $x(t)$ is real signal → verify, $X(f) = X^*(-f)$,

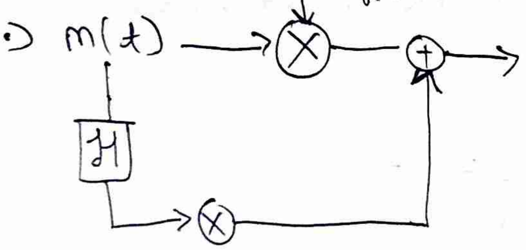
• $x(t) = m_+(t)[\cos 2\pi f_c t + j\sin 2\pi f_c t]$
$m_-(t)[\cos 2\pi f_c t - j\sin 2\pi f_c t]$

$= \underset{\substack{\downarrow \\ \text{In phase component}}}{m(t)\cos 2\pi f_c t} - \underset{\substack{\downarrow \\ \text{Quadrature}}}{\overbrace{(-j m_+(t) + j m_-(t))}^{m_h(t)}} \sin 2\pi f_c t$

↳ $m_h(t) = \mathcal{H}(m(t))$ → frequency response $-j\,\text{sign}(f)$

In frequency $-jM_+(f) + jM_-(f)$



• 

• find inverse transform of $-j\,\text{sign}(f) \longleftrightarrow \frac{1}{\pi t}$

~~We can see to send our message comp!~~

- Now analyse $(m(t)+A_c)\cos 2\pi f_c t - m_h(t)\sin 2\pi f_c t$

$$= e(t)\left[\cos(2\pi f_c t + \phi(t))\right]$$

$$e(t) = \sqrt{(m(t)+A_c)^2 + m_h^2(t)} \simeq A_c + m(t)$$

→ hilbert component being negligible ~~makes us~~ use envolope method seen previously ~~to demodulate~~ the signal ↓ In SSB

→ Check

- $m(t)$ is bandlimited so $m_h^2(t)$ is not unbounded

- The demodulation circuit is the same as one we saw for Down conversion, we get $m(t)$ in the inphase arm, we are just sending a different looking signal.

QAM → Quadrature Amplitude modulation

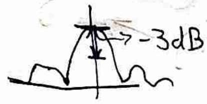
↳ 2 AM signals which are orthogonal, this is widely used

↳ Amplitude shift Keying → Write from Sensor

Assignment 1

- ↳ find out why the spectrum was comming up with imaginary?
  ↳ indexing of fft was wrong. you had to shift the time domain signal too

- 3dB bandwidth → log plot of energy Spectral density, look at 3dB below the peak amplitude



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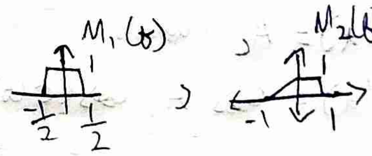
Assignment 2

(I) Upconversion

Generate baseband signal by taking iFFt of



Upconvert to $f_c = 10\,Hz$

plot the Converted time and frequency

2) Down conversion

Downconvert your upconverted signal.

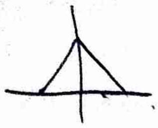
• QAM → $x(t) = A_c[m_I(t)\cos 2\pi f_c t - m_Q(t)\sin 2\pi f_c t]$

Hilbert transform

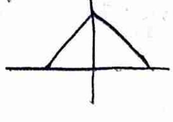
$H(x(t)) = x(t) * \frac{1}{\pi t}$

$X_h(f) = X(f)(-j\,\text{Sign}(f))$

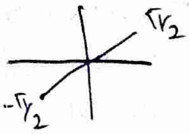
$|X(f)|$



$|X_h(f)| =$

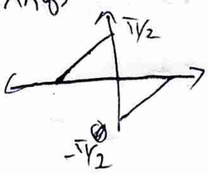


$\measuredangle X(f)$



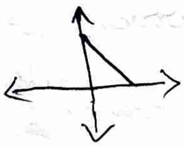
→

$\measuredangle X_h(f)$



• Spectrum of $x(t) + j x_h(t)$ → analytic representation of a signal

$= X(f) + X(f)\,\text{Sign}(f)$ →



• this is a technique to remove -ve freq values in spectrum

• After this you upconvert it,

• Section 2.8.3 about complex baseband is interesting
↳ in text book

Angle Modulation

• We will work on $\phi(t)$, and we have 2 options

Phase Modulation (PM) $\boxed{\phi(t) = k_p m(t)}$ → deviation constant

freq Modulation (FM) → $\boxed{f(t) = \frac{1}{2\pi}\frac{d\phi}{dt} = k_f m(t)}$

↳ $\boxed{\phi(t) = 2\pi k_f \int_0^t m(\tau)\, d\tau}$

if $e(t) = A_c$ (constant)

$m_I(t) = A_c \cos(\phi(t))$

$m_Q(t) = A_c \sin(\phi(t))$

• ~~After~~ If LO is not synchronize we get

$\cos\theta \cos\phi(t) + \sin\theta \sin\phi(t)$ in the I arm we receive $\cos(\phi(t) - \theta)$ and $\sin(\phi(t) - \theta)$ ← Phase delay

we can do $\cos^{-1}$ it to recover phase ⇒ which contains our message

- this is good for PM, but ~~...~~ how do you recover for F.M

Assumption $m(t)$ has no DC component

- you derivate $\frac{1}{2\pi}\frac{d}{dt}(\phi(t) - \theta) \Rightarrow \frac{1}{2\pi}\frac{d}{dt}(\phi(t))$ → so θ just goes away.

so we prefer FM

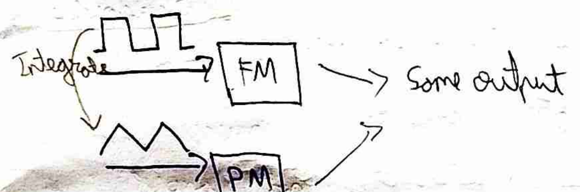
- Circuits for Both FM and PM ~~...~~ modulations work for each other

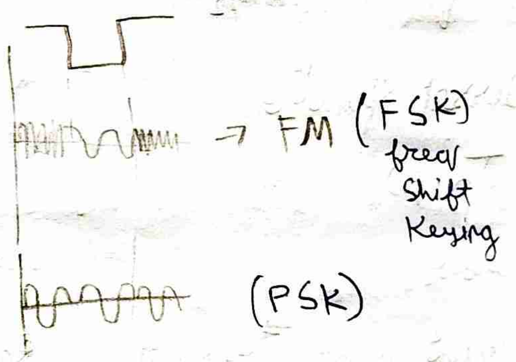
$m(t) \rightarrow \boxed{FM} \rightarrow X_{FM}(t)$

$m(t) \rightarrow \boxed{\frac{d}{dt}} \rightarrow \boxed{FM} \rightarrow X_{PM}(t)$ → $K_p = 2\pi K_f$

$m(t) \rightarrow \boxed{\int dt} \rightarrow \boxed{PM} \rightarrow X_{FM}$

- What happens to the spectrum after modulation?





- Satellite ~~using~~ use FSK, digital communication ~~uses~~ uses PSK

- Modulation index $\beta = \cancel{k_f \max(m(t))}\ \frac{k_f \max(m(t))}{B} = \frac{\Delta f}{B}$

multiply

B → Bandwidth of the message

Δf → max freq division

B → Bandwidth

For FM

$= k_p \max(m(t))$ For PM

Max deviation of freq from the carrier freq

• looking at spectrum of $\boxed{FM}$, PM is the Same

• Narrow band FM : → Δf is very small, $\phi(t) << 1$

$$A_c\left[\underset{\approx 1}{\cos\phi(t)}\cos 2\pi f_c t - \underset{\approx \phi(t)}{\sin\phi(t)}\sin 2\pi f_c t\right]$$

this is again similar to A.M

$X_{FM}(f)$ has bandwidth as $2B$, we assume band width is same even after integration

• We have to use spectrum of $Re\left\{e^{j2\pi f_c t + j2\pi k_f \int_0^t m(\tau)\,d\tau}\right\}$

Say $m(t) = \cancel{\cos(2\pi f_m t)}\ \cos(2\pi f_m t)$ → Band width $B = f_m$

we have $\beta = \dfrac{k_f \times 1}{B}$ (modulation index) →

$e^{j\beta \sin 2\pi f_m t}$

• This signal is periodic

$\therefore C_n = \dfrac{1}{2\pi}\displaystyle\int_{-\pi/2}^{\pi/2} e^{j(\beta \sin \omega - n\omega)t}\,dt$ $\qquad 2\pi f_m = \omega$

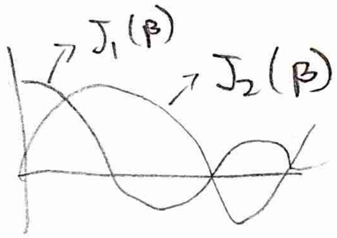
↓

Bessel function

$J_n(\beta)$ →

↓

order



• What we are transmitting $Re\left\{e^{j2\pi f_c t}\sum J_n(\beta)e^{j2\pi f_m t}\right\}$

$$\boxed{= \sum J_n(\beta)\cos(2\pi f_c t + n2\pi f_m t)}$$

Assignment-2, Q7

• Message Value goes to ∞

• do we want $\beta << 1$

• Do we want lesser bandwidth for FM because compression

•) New class — fourier series of

$$c_n = \frac{1}{1/f_m} \int e^{j\beta \sin 2\pi f_m t} \, e^{-j 2\pi n f_m t} \, dt \qquad V = 2\pi f_m$$

$$\Rightarrow J_n(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin V - nV)} \, dV$$ → n^{th} order, 1st kind bessel function

•) This doesn't depend on f_m? wrong these are just the coefficient of fourier series, finally we have $\sum_{n=-\infty}^{\infty} J_n(\beta) e^{j 2\pi f_m n}$

•) $J_n(\beta) \approx \frac{(\beta/2)^n}{n!}$ for $\beta << 1$

•) This decays very fast, so let us try to get the 99% bandwidth of the signal from the coefficients

$$99\% \text{ BW} = \sum_{n=-K}^{K} J_n^2(\beta)$$

$$= J_0^2(\beta) + 2\sum_{n=1}^{K} J_n^2(\beta)$$ → there is a symmetry

•) Bessel functions appear in a lot of places → drum surface vibrations, and a lot of places → find

•) ~~Bandwidth $\approx 2(K+1)$~~

•) Carson's Rule | $K \approx \beta + 1$ →

•) ∴ Bandwidth $\approx$ ~~$2(K+1)$~~ $2(\beta+1) f_m$ → Bandwidth of message

•) So there will be some distortions, we do not send $e^{j\beta \sin f_m t}$ but $\sum_{n=-K}^{K} e^{j 2\pi n f_m}$

•) How do you modulate FM signals → Voltage Controlled Oscillator (VCO)

•) say $f(t) = K_V V(t) + f_c$, we get $\cos\left(2\pi f_c t + 2\pi K_V \int_0^t V(\tau) d\tau\right)$

derivate it

•) How do you ~~de~~ demodulate the signal, you differentiate it to get the freq outside

we get $A_c \sin(\ldots) \times (2\pi k_f m(t) + 2\pi f_c)$

Δf the frequency deviation should not be large to ~~ded~~ demodulate

↳ this is now similar to AM, and ~~so~~ we can use that ~~to~~ system

•) For P-M the analysis is still the same, we get a shift in $J_n(\beta)$

•) But ~~the~~ bandwidth changes

$BW_{FM} \simeq 2(\Delta f + f_m)$ | $BW_{PM} \simeq 2(\Delta\phi + 1) f_m$ → Check

•) For the same message FM has a ~~less~~ lesser bandwidth

•) FM or PM is spectrally inefficiency ~~o~~ compared to ~~A~~ AM, but in the power efficiency it is almost 100% in FM, this becomes significant when noise ~~o~~ is introduced (We want high SNR)

New class

$\beta_{PM} = k_p \max|m(t)|$, $\beta_{FM} = \dfrac{k_f \max|m(t)|}{B}$

•) We want a better demodulator than a diffrentiator for FM

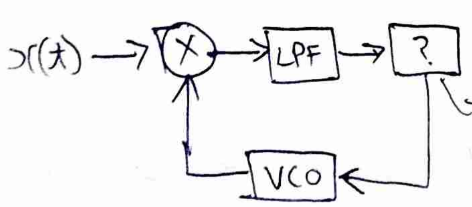
Phase Locked Loops PLL

$x(t) = A_c \cos(2\pi f_c t + \phi(t)) \longrightarrow (\times) \longrightarrow \dfrac{A_c A_v}{2}\left[\cancel{\sin(4\pi f_c t + \phi(t) + \phi_v(t))} + \sin(\phi(t) - \phi_v(t))\right]$

removed from LPF

VCO: $-A_v \sin(2\pi f_c t + \phi_v(t))$

↳ see last page



↳ we drive the error to zero

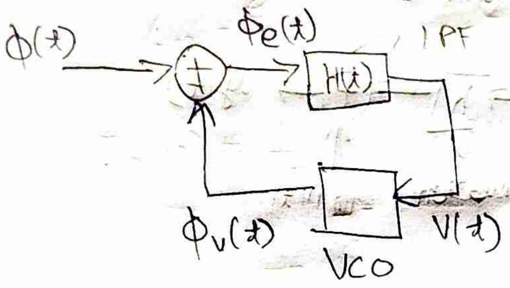
↳ How, we have $\dfrac{A_c A_v}{2}(\sin(\phi(t) - \phi_v(t))$

if this goes to zero then, $\phi_v(t) \simeq \phi(t)$ our ~~ded~~ demodulated signal

our message is what is present in $\phi_v(t)$ → Next page

→ first order PLL

•) Assumption $\approx \frac{A_c A_v}{2}\left[\phi(t) - \phi_v(t)\right]$ if error is small



as $f(t) = f_c + k_v V(t)$ for the VCO

•) $\phi_e(t) + \phi_v(t) = \phi(t) \Rightarrow \frac{d\phi_e}{dt} + K_v V(t) = \frac{d}{dt}\phi(t)$

$$\Rightarrow \frac{d\phi_e(t)}{dt} + 2\pi K_v h(t) * \phi_e(t) = \frac{d\phi(t)}{dt}$$

↓ Fourier transform

$$j2\pi f\, \Phi_e(f) + 2\pi K_v H(f) \times \Phi_e(f) = 2\pi f j \Phi(f)$$

$$\Rightarrow \Phi_e(f) = \frac{\Phi(f)}{1 + \frac{K_v H(f)}{jf}}$$

•) $V(f) = H(f)\Phi_e(f)$

$$= \frac{H(f)}{1 + \frac{K_v}{jf}H(f)}\Phi(f)$$

assume $|H(f)| >> 1$ (put an amplifier)

↓

inside the bandwidth of the message → So we can use a LPF insted of an amplifier, Gain of LPF should be high

$$\therefore V(f) = \frac{jf}{K_v}\Phi(f)$$

$$V(t) = \frac{1}{2\pi K_v}\frac{d\phi(t)}{dt} = \frac{K_f}{K_v}m(t)$$ ⇒ $V(t)$ is proportional to our message

↳ this is a PI controller, VCO is the integrator block

•) Pg128 of text book has a lot more content on PLL

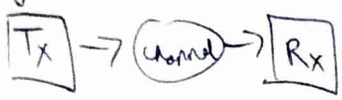
New class

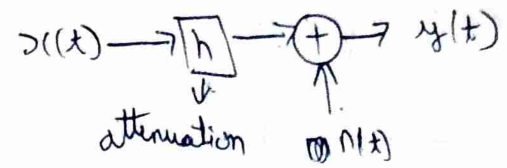
•) PLL is unstable → the message should be bandlimited

•) If our message has a DC component, the integrator causes the system to become unstable

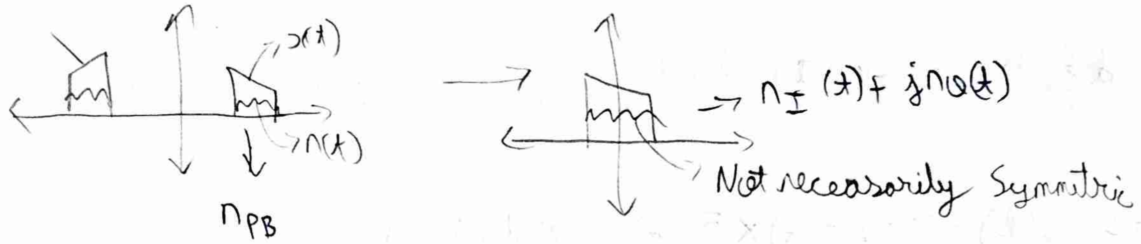
- Would you pick AM or FM?
- Let us look at Additive Noise in channels

$y = hx + n$





- There is noise in baseband and passband, but the passband noise is real, but baseband noise ~~~~ is not necessarily real



- Our LPF determines the quality of noise effect.
- In AM, if

			Noise	
	AM	FM	AM	FM
$M_{demod}\ I(t) =$	$m(t)$	$\cos\phi(t)$	$m(t) + n_I(t)$	$\cos\phi(t) + n_I$
$M_{demod}\ Q(t) =$	0	$\sin\phi(t)$	$n_Q(t)$	$\sin\phi(t) + n_Q$

- demodulation is simpler for FM → take derivative

at the reciever lets say we have

$$\cos\phi(t) + n_I(t) + j(\sin\phi(t) + n_Q(t))$$

↓ combine terms

we get $e^{j\phi(t)} + r(t)\, e^{j\theta(t)}$ → - we project this signal into the unit circle in the ~~~~ complex plane,

our message $e^{j\phi(t)}$ will always be found there while noise can be anywhere.

- We do this as our message signal is contained in the phase.

New Class

- FM has a better power efficiency → which makes it ~~more~~ robust to noise

- $y(t) = h\, x(t) + n_{RF}(t)$, with $y(t)$: Rx, h: attenuation, $x(t)$: Tx, $n_{RF}(t)$: noise, where $n_{RF}(t) \to n_I(t) + j\, n_Q(t)$

Random Variables

$X(t) \to$ random variable

↳ Sample Space, F → event space, Probability Measure

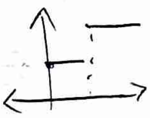
Ω : $\{H, T\}$; F : 2^{Ω}; P ↳ P.D.F

$X : \Omega \to \mathbb{R}^n$

- Borel Set → Our message is a random variable, but ~~as~~ Ω is uncountably ∞ so you use this to ~~pick~~ model probability

- PDF for a few random variable may not even exist, but CDF exists
  ↳ HW

- CDF → monotonic, Right Continuous



$F_X(x) = P(X \leq x)$

- $\mathbb{E}[g(X)] = \int g(x) f_X(x)\, dx$

  $\mathrm{Var}(X) = \mathbb{E}\{(X - E[X])^2\}$

  $\mathrm{Cov}(X, Y) = \mathbb{E}\{(X - \mathbb{E}(X))(Y - E[Y])\}$ — ~~Cross~~ Cross Covariance; Cor → see

If the Random variable is Complex you will do conjugate

$\mathrm{Var}(X) = E[(X - E[X])^*(X - E[X])^*]$ → this is real

$\mathrm{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))^*]$ → this can be complex

- Bernoulli → $X \in \{x_0, x_1\}$, $x_0 \to p$, $x_1 \to 1-p$; $Be(p)$

- Uniform → $X \sim U[a, b]$, $f_X(x) = \begin{cases} \frac{1}{b-a} \\ 0 \end{cases}$

• Gaussian $\rightarrow X \sim \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

• $f_{XY}(x,y) = f_X(x) f_Y(y) \rightarrow$ means Independent

$E[X.Y] = E[X]E[Y] \rightarrow$ Uncorrelated

• Gaussian R.V are ~~Uncorr~~ Uncorelated and Independence

$Z = \begin{bmatrix} X \\ Y \end{bmatrix}$, $\frac{1}{2\pi \det(\Sigma)} e^{-\frac{1}{2}(Z-\mu)^H \Sigma^{-1} (Z-\mu)}$

If $X_{n\times 1}$ $\rightarrow$ π^n

$\Sigma = \begin{bmatrix} C_{ij} = Cov(X_i, X_j) \end{bmatrix}$

$\begin{bmatrix} n_I(t) \\ n_Q(t) \end{bmatrix} = N$, $n_I \rightarrow N(0,\sigma_I^2)$, $n_Q \rightarrow N(0,\sigma_Q^2)$ $\longrightarrow \frac{1}{2\pi (\det\Sigma)^{1/2}} e^{-\frac{1}{2}\underbrace{\left(\begin{bmatrix} n_I \\ n_Q \end{bmatrix}^T \Sigma^{-1} \begin{bmatrix} n_I \\ n_Q \end{bmatrix}\right)}_{\frac{n_I^2}{\sigma_I^2} + \frac{n_Q^2}{\sigma_Q^2}}}$, $(\det\Sigma)^{1/2} \rightarrow \sigma_I \sigma_Q$

They are independent $\quad \Sigma = \begin{bmatrix} \sigma_I^2 & 0 \\ 0 & \sigma_Q^2 \end{bmatrix}$

• If You transform N to polar Coordinates what happens to the distribution of $\begin{bmatrix} r \\ \theta \end{bmatrix}$ from $\begin{bmatrix} n_I \\ n_Q \end{bmatrix}$

~~$R = \sqrt{n_I^2 + n_Q^2}$~~ $R = \sqrt{N_I^2 + N_Q^2}$, $\theta = \tan^{-1}\left(\frac{N_Q}{N_I}\right)$

• $Y = g(X)$, $f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'()|} = \frac{f_X(g^{-1}(y))}{\det(J)}$ $\rightarrow$ for multivariable

$N_I = R\cos\theta$

$N_Q = R\sin\theta$, find Jacobian of this

• HW, find $f_{R,\Theta}(r,\theta) = ?$

New class

• Random Vector

$\hookrightarrow$ Co-varience matrix

If $X \in \mathbb{R}^n$, $X \sim N(\mu, \Sigma)$

$$= \frac{1}{(2\pi)^{n/2} \det(\Sigma)} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

If $X \in \mathbb{C}^n$ $$= \frac{1}{(2\pi)^n \det(\Sigma)} e^{-\frac{1}{2}(x-\mu)^H \Sigma^{-1}(x-\mu)}$$

• If Σ is diagonal matrix $\rightarrow$ distribution is independent so multiply the marginals to get joint P.D.F

• Yesterday we were doing $\begin{bmatrix} n_I \\ n_Q \end{bmatrix} \xrightarrow{X \to Y} \begin{bmatrix} r \\ \theta \end{bmatrix}$

$\hookrightarrow$ Pg 218 in textbook

• $$J = \begin{bmatrix} \frac{\partial r}{\partial n_I} & \frac{\partial r}{\partial n_Q} \\ \frac{\partial \theta}{\partial n_I} & \frac{\partial \theta}{\partial n_Q} \end{bmatrix} = \begin{bmatrix} \frac{n_I}{\sqrt{n_I^2 + n_Q^2}} & \frac{n_Q}{\sqrt{n_I^2 + n_Q^2}} \\ \cancel{\frac{-n_Q}{\sqrt{n_I^2+n_Q^2}}} \;\; \frac{-n_Q}{n_I^2 + n_Q^2} & \frac{n_I}{n_I^2 + n_Q^2} \end{bmatrix}$$

• you have to divide by det(J) for your distribution

$\hookrightarrow$ If you are doing a transformation

$\det(J) = \frac{1}{r}$

• $$f_{r,\theta}(r,\theta) = \frac{r}{2\pi\sigma^2} e^{-\frac{1}{2}\left(\frac{r^2}{\sigma^2}\right)}$$

• $$f_r(r) = \int_{-\pi}^{\pi} (\dots)\, d\theta = \frac{r}{\sigma^2} e^{-\frac{1}{2}\left(\frac{r}{\sigma}\right)^2} \longrightarrow \text{Rayleigh distribution}$$

• $$f_\theta = \frac{f_{r,\theta}}{f_r} = \frac{1}{2\pi} \rightarrow$$ they are uniform, $\Theta \sim \text{Unif}[-\pi, \pi]$

$R \sim \frac{r}{\sigma^2} e^{-\left(\frac{r}{\sigma}\right)^2}$

$\hookrightarrow$ distribution in this direction

• $$[1 \;\; 0] \begin{bmatrix} n_I \\ n_Q \end{bmatrix} \rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{n_I^2}{2\sigma^2}}$$

• What about a distribution in any other direction $\rightarrow$ we get the same distribution $\rightarrow$ Symmetric around θ

- They are called, Circular Symmetric Complex Gaussian
- If $\sigma_Q \neq \sigma_I$ it is not symmetric
- $y(t) = h\,x(t) + n(t)$

our $n(t)$ is a random process $X_1, X_2 \cdots X_n \rightarrow$ a sequence
but we have Continious time random process, ∴ $X(t)$

- For any random variable we look at its statistics (Mean etc)
- For a random process? $\rightarrow X_{t_1} - X_{t_2}$

different time same statistic $\rightarrow$ strict sense Stationary process

In real life Wide sense stationary - (WSS)

1) mean is stationary $m_X(t) = \mathbb{E}\{X(t)\} = m_X$ (Constant)

This is a signal deterministic

Auto correlation

2) $R_X(t_1, t_2) \rightarrow$ Correlation of random process $= \mathbb{E}[X_1(t_1)X^*(t_2)]$
$\hookrightarrow$ auto correlation

$$= \int x_{t_1} x^*_{t_2} f(x_{t_1}, x_{t_2})\, dt_1\, dt_2$$

depends on $t_1 - t_2$ not t_1 or t_2

if this $= R_X(t_1 - t_2) \rightarrow$ (WSS)

$\hookrightarrow$ Samples seperated by Same amount of time have a similar distribution

- $X(t)$ WSS $\rightarrow m(t)$, $\hookrightarrow R(\tau)$ we can see $R(-\tau) = R^*(\tau)$
$\hookrightarrow$ will have real spectrum

- Example of random process $X(t) = A_c \cos(2\pi f_c t + \Theta)$, $\Theta \sim$ Unif $[-\pi, \pi]$

$$m_X(t) = \mathbb{E}[X(t)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} A_c \cos(2\pi f_c t + \theta)\, d\theta = 0 \rightarrow \text{it is stationary}$$

$\cdot)\ R_x(t_1, t_2) = \int \cancel{x(t_1)\, x^*(t_2)} \quad \mathbb{E}\{X(t_1) X^*(t_2)\}$

$$= \int A_c^{*} \cos(2\pi f_c t_1 + \theta)\, A_c \cos(2\pi f_c t_2 + \theta)\, dx_{t_1}\, dx_{t_2}$$

here θ is random | but it is fixed (doesn't change with time)

↳ this is WSS → How?

New class

↳ for analog systems we have continuous time random process

$$Y(t) = h\, X(t) + N(t)$$

$Y(t)$ ↓ R_x, $X(t)$ ↓ T_x, $N(t)$ ↓ AWGN

$\cdot)$ WSS $\doteq$ $m_x(t) = m_x$ | $R_{xx} = R_{xx}(t_1 - t_2)$ ↗ lag or delay

$\cdot)$ Example from last class

$$X(t) = A_c \cos(2\pi f_c t + \theta)$$

↳ θ Unif $[-\pi, \pi]$

$= g(\theta, t)$ → θ is fixed after being picked randomly but different t gives a new value → this is a random process

$\cdot)\ \mathbb{E}\{x(t)\} = ?$

↳ $\cdot\ E\{g(\theta)\} = \int g(\theta)\, f_\theta(\tau)\, d\tau \qquad \phi = g(\theta)$

$= \int \phi\, f_\phi\, d\tau$ → Prove this

$$f_\phi = \frac{f_\theta(g^{-1}(\phi))}{g'}$$

$f_\theta(\theta)$?

from last page

•) $E\{X(t)\} = \int_{-\pi}^{\pi} A_c \cos(2\pi f_c t + \theta) \frac{d\theta}{2\pi} = 0$

•) $R_{XX}\{X(t)\} = \iint A_c \cos(2\pi f_c t_1 + \theta)\, A_c \cos(2\pi f_c t_2 + \theta)\, f_X(x_{t_1}, x_{t_2})\, dx_{t_1}\, dx_{t_2}$

•) We are taking ensemble average ↑, the process is ergodic

↳ try transforming to $g(\theta)h(\theta)$ and integrate over θ

•) We transform and integrate over θ

we get $R_{XX}(t) = \frac{A_c^2}{2} \cos(2\pi f_c (t_1 - t_2))$

•) Say during transmission $X(t) = M(t) \cos 2\pi f_c t$

Say it is WSS

$E(X(t)) = E\{M(t)\} \cos 2\pi f_c t$ → Periodic

•) But at the reciever $X(t) = M(t)$ → which is again WSS

↳ We see blocks in comsys change the stationary state

what is R_{XX} of this ?

$R_{XX}(t) = E\{M(t) \cos 2\pi f_c t_1,\ M^*(t) \cos 2\pi f_c t_2\}$

$= E\{M(t_1) M^*(t_2)\} \{\cos(2\pi f_c (t_1 - t_2)) + \cos(2\pi f_c (t_1 + t_2))\}$

↳ How do you get this in a computer

↳ How to simulate

mean and Rx is periodic

↑

Q $\boxed{\text{Cyclostationary}}$ → the above this is periodic

•) Sensors use this (CSMA) to see if a band is occupied or not

- $X(t)$ is periodic with $\frac{1}{f_c}$

- How to say it is periodic $R_{XX}(t_1+T_1, t_2) = R_{XX}(t_1, t_2)$
$= R_{XX}(t_1, t_2+T_2)$

$T_1 = T_2$ for cyclostationary

- What happens to the statistic of a random process after going through ~~a~~ LTI system

$X(t) \to \boxed{LTI} \to Y(t)$ $\quad$ $Y(t) = h(t) \circledast X(t)$

$h(t)$

$\mathbb{E}\{Y(t)\} = \mathbb{E}\left\{\int X(\tau)\, h(t-\tau)\, d\tau\right\}$ ← looking at mean first

- we can swap ~~integrals~~ expectation if integrals are defined

$\Rightarrow \int \mathbb{E}(x(\tau))\, h(t-\tau)\, d\tau$

Not random

$t-\tau$ doesn't matter

$\mathbb{E}\{Y(t)\} = m_X(t) \int_{-\infty}^{\infty} h(\tau)\, d\tau$ → if system is unstable we cannot say anything

→ not same as $\int_{-\infty}^{\infty} |h(t)| < \infty$ → $H(0)$ ($f=0$)

if ~~LTI~~ system is not stable. our expectation becomes ∞.

- ~~R_{XY}~~ defining cross ~~correla~~ correlation

~~R_{XY}~~ $= \mathbb{E}\{X(t_1)\, Y^*(t_2)\}$

$R_{XY}(t_1, t_2)$

- if X, Y ~~are~~ are WSS and if $R_{XY}(t_1 - t_2)$ → τ

they are jointly WSS

- We again have a property ~~R_{XX}~~ $R_{XY}(\tau) = R^*_{YX}(-\tau)$

→ X and Y are even and real then $R_{XY} = R_{YX}$

New class

- $R_{XY} = \mathbb{E}\{X(t+\tau)\,Y(t)\}$ → if X, Y are real

$R_{YX} = \mathbb{E}\{Y(t+\tau)\,X(t)\}$

- when are the above things equal → When again?
- how do you say two random ~~process~~ Variables are same
  1. ↳ Strict ~~same~~ equality → basically duplicate
  2. ↳ Almost sure $X(w) = Y(w)$ when $P(w) \neq 0$
  3. ↳ Same in distribution X and Y can ~~repr~~ represent multiple things but their distribution is same.

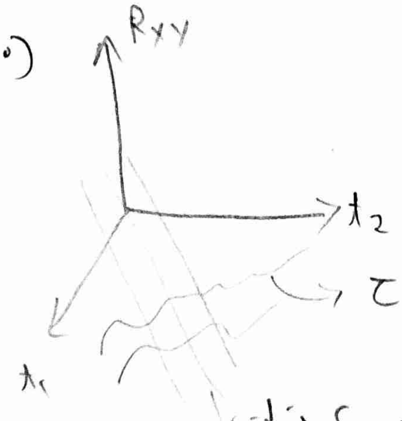
→ we are taking about 2,3 in our case of R_{XY}

$Y(t+\tau)\,X(t) \Rightarrow Y(-t-\tau)\,X(-t)$

$-t-\tau \to v$, $-t \to v+\tau$ $\Rightarrow Y(v)\,X(v+\tau)$

⇓ equal to above first thing

- X was τ ahead of Y, Same here after flipping

- 

// Check again

↳ τ line, we only see a 1-D curve instead of 3-D

↳ it is Same along here (these lines)

- $R_{xx}(\tau) = \mathbb{E}\{X(t+\tau)X(t)\}$ → for real

$R_{xx}(-\tau) = \mathbb{E}\{X(t-\tau)X(t)\}$

- $X(t) \to \boxed{LTI} \to Y(t)$
  $h(t)$

$$R_{YX}(t_1,t_2) = \mathbb{E}\{X^*(t_2)\underbrace{\int X(t_1-\tau)h(\tau)d\tau}_{Y(t_1)}\}$$

$$= \int \mathbb{E}\{X(t_1-\tau)X^*(t_2)\}h(\tau)d\tau$$

$\cancel{R_{XX}(\ )*h(\ )}$ $R_{YX} = R_{XX}(v) * h(v)$

$= R^*_{YX}(-\tau)$

- $R_{XY} = R^*_{XX}(-v) \circledast h(-v)$ → statistics of random process are also convolved with $h(t)$ → Because of linearity

$= R_{XX}(v) \circledast h(-v)$

- $R_{YY}(\tau) = R_{XX} \ * h(v) * h^*(-v)$

New class

- $X(t), Y(t)$ → are WSS and $R_{XY}(t_1-t_2) = R_{XY}(\tau) = R^*_{YX}(-\tau)$ then X, Y are jointly WSS

- ~~R_{XY}~~ $R_{YX} = R_{XX}(\tau) * h(\tau)$ → this ~~does~~ doesn't tell us $Y(t)$ is WSS

- $R_{XY} = R_{XX}(\tau) \circledast h^*(-\tau)$

- $R_{YY} = \{Y(t_1)Y^*(t_2)\}$

$$= \mathbb{E}\int X(t_1-\tau)h(\tau)d\tau\, Y^*(t_2)$$

$$= \int R_{XY}(t_1-\tau-t_2)h(\tau)\cdot d\tau \qquad \text{take } t_1-t_2 = V$$

$$= \int R_{XY}(V-\tau)h(\tau)d\tau$$

$$R_{YY} = R_{XY}(V) \circledast h(V) = R_{XX}(V) * h^*(-V) * h(V)$$

- $R_{YY} = R_{XX}(v) * h^*(-v) * h(v)$

- Looking at the spectrum of a random process

- Say $X \in \mathbb{R}^n \sim N(\mu, \Sigma)$ gaussian random vector

  $\Sigma \rightarrow$ Co-varience

- $Y = F X$

  $F \rightarrow$ DFT Matrix, Y is also gaussion $\sim N(F\mu, F\Sigma F^H)$

  If $\Sigma = \sigma^2 I \rightarrow$ i.i.d

  then the distribution of Y is also gaussion

  F is a unitary matrix (Just rotates)

- Instantanious spectrum is just random.
- $\mathbb{E}\{|X(f)|^2\} \rightarrow$ PSD (Power Spectral Density)

  $= \mathcal{F}\{R_{XX}(\tau)\}$ $\rightarrow$ Wiener–Khinchin Theorem

  $\hookrightarrow$ if it is WSS

- ~~Oscillators~~ Oscilloscopes take and show PSD $\rightarrow$ they cannot show phase of Fourier transform.

- When does the fourier transform gurantee real and positive output? as we are doing $\mathbb{E}\{|\cdots|^2\}$

  real $\rightarrow$ Conjugate Symmetric

- $|X(f)|^2 = X(f)\, X^*(f)$

  $= \int X_1(t_1)\, e^{-j2\pi f t_1}\, dt_1 \int X_2^*(t_2)\, e^{+j2\pi f t_2}$

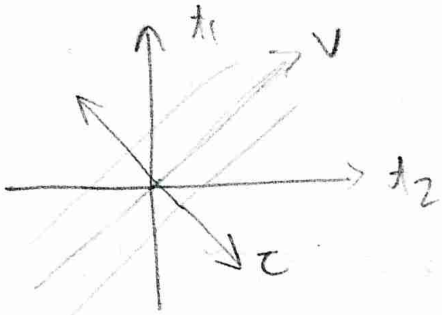
  $= \int x(t_1)\, x^*(t_2)\, e^{-j2\pi f(t_1 - t_2)}$

$E\{|X(f)|^2\} = \iint R_{XX}(\tau)\, e^{-2\pi f j}\, dt_1\, dt_2$

•) $x(t)$ $\quad x^*(-t)$ → $x(t) * x^*(-t)$ will always have a non negative transform

$\downarrow \qquad \downarrow$

$X(f) \qquad X^*(f)$



transform from t_1, t_2 to τ, V

$\tau = t_1 - t_2 \qquad V = t_1 + t_2$

$\tau \perp V$

we get

$$\begin{bmatrix} \tau \\ V \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}}_{J} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$= \iint R_{xx}(\tau) e^{j2\pi f t} \frac{d\tau \cdot dV}{2}$$ → ~~wed~~ what about limits?

$t_1 - t_2$, $\lim_{t_1, t_2 \to \infty}$ is what?

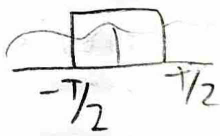
•) But here we get ∞ in the integration

•) So we normalize

$\lim_{T \to \infty} \frac{1}{T} \mathbb{E}\{|X_T(f)|^2\}$, where $X_T(f) = \mathcal{F}\{X(t) \cdot \text{Rect}(\frac{t}{T})\}$

↓

This is our definition of PSD



•) $N(t) \sim N(0, \sigma^2)$

$m_N(t) = 0$, $R_{NN}(t_1, t_2) = \delta(\tau)$ $t_1 \neq t_2$ $\mathbb{E} = 0$

Quiz 1

Q1 $\frac{1}{2\pi} \frac{d\phi(t)}{dt} = f(t) = \phi(t)$

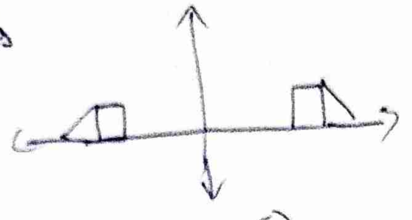
Q2 ⇒ Can have a ~~highpass~~ highpass filter at a higher freq

↳ Consider carrier freq as zero → modulate ~~of~~ FM signal with that assumption.

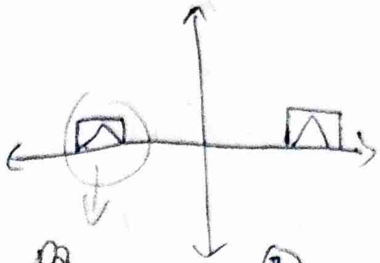
$x(t) = (0 + k_f \int_0^{\tau} m(\tau) d\tau)$ → then transport $x(t) + j x_h(t)$

hilbert transform

Q3 Signals



Vs



Both have same bandwidth efficiency.
In real life we use (2), because hardward is simple. Only one oscillator

Q5 In signal power → if signals are orthogonal you can just add their ~~phase~~ power

Q6 $Y(t) = M(t)\cos(2\pi f_c t + \Theta) + N(t)$

$\underbrace{M(t) \to \text{WSS},\ \cos \to \text{we know it is WSS},\ \Theta \to \text{Unif}}_{R_{xx}(\tau)}$ $\quad \underbrace{N(t) \to \text{i.i.d } N(0, \sigma^2)}_{\text{also WSS}}$

they are independent so you can add them

∴ ~~Y(t) E R~~

∴ $Y(t)$ is WSS

- Ideal LPF, $h(t) * Y(t)\cos 2\pi f_c t$ is still WSS

- New class

Why is Autocorrelation a Power signal → It is a 2D signal only for a WSS signal ~~which~~ it is a 1D signal

- Auto correlation function can never be zero → implies energy is zero.

- Is there a signal whose $\int R_{xx}(\tau) = 0$ → $\cos(2\pi f_c \tau)$ → in the limit of ∞

- We look at auto correlation in a ~~window~~ window, for defining PSD

we have $\lim_{T \to \infty} \frac{1}{T} E\{|X_T(f)|^2\} = \int_{-T/2}^{T/2}\int_{-T/2}^{T/2} R_{xx}(t_1 - t_2)\, e^{-j2\pi f_c(t_1 - t_2)}\, dt_1\, dt_2$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-T}^{T}\int_{-T+|\tau|}^{T-|\tau|} R_{X_T X_T}(\tau)\, e^{-j2\pi f\tau}\, \frac{dv\,d\tau}{2}$$

→ Not WSS, only when $T \to \infty$

→ We have just rotated it

$\tau = t_1 - t_2$

$v = t_1 + t_2$

New class

- Wiener ~~Kin~~ Khinchin Theorem

PSD of $X(t) = \mathcal{F}\{R_{xx}(\tau)\}$

will it work if X is SSS ⇒ Yes (Because it is still WSS)

SSS says their $(X_1(t_1),\ \cancel{X_2(t_2)}\ X_1(t_1 - d))$ have the same distribution

we had then $\displaystyle \lim_{T\to\infty} E\left\{ \frac{1}{T} \int_{-T/2}^{T/2}\int_{-T/2}^{T/2} X_1(t_1) e^{-j2\pi f_c t_1}\, X_2^*(t_2) e^{j2\pi f_c t_2}\, dt_1\, dt_2 \right.$

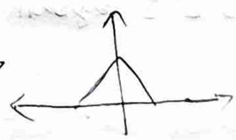
→ limits come from rect(t/T)

but $X_1(t) = X(t)\ \text{rect}(t/T)$ ⇒ this is not WSS

i) $X_1(t)$, and R_{xx} need to satisfy dirichlet conditions → then we can take the expectation inside.

$$\Rightarrow \lim_{T\to\infty} \frac{1}{T}^{?} \int_{-T}^{T}\int_{-T+|\tau|}^{T-|\tau|} R_{xx}(\tau)\, e^{-j2\pi f\tau}\, \frac{dv\,d\tau}{2}$$

$$= \lim_{T\to\infty} \frac{1}{T}^{?} \int_{-T}^{T} R_{xx}(\tau)\, e^{-j2\pi f\tau}\, d\tau \times \left(\frac{2T - 2|\tau|}{2T}\right) \to 1 - \frac{|\tau|}{T} \to$$



$\downarrow$

$\frac{1}{T}\ \text{rect}\left(\frac{\tau}{T/2}\right) \circledast \text{rect}\left(\frac{\tau}{T/2}\right)$

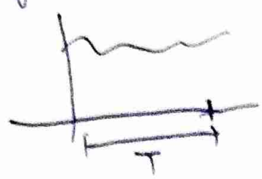
Check

↳ fourier transform

$\lim_{T\to\infty} R(f) * T\,\text{sinc}^2(fT)$

as $T \to \infty$, this tends to $\delta(f)$, we get $R(f) * \delta(f) \Rightarrow R(f)$

·) How to find PSD & IRL

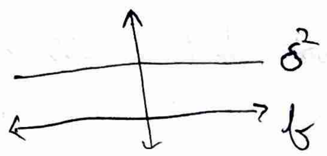
$X(t)$: 

take $\int x(t)\, x^*(t-\tau)$ in different windows and correlate with it self with the Same delay → then average over
↓
This is the expectation

·) Small window may not capture the low freq signals in that window, Large windows are never a problem → you do not need to go beyond the lowest freq.

·) AWGN Model

$Y(t) = X(t) + N(t)$, $N \sim \mathcal{N}(0, \sigma^2)$

$\therefore R_{NN}(\tau) = \sigma^2 \delta(\tau) \rightarrow \mathcal{F} \rightarrow$ 

it is present in all frequencies → white

·) But this has ∞ Power → SNR=0

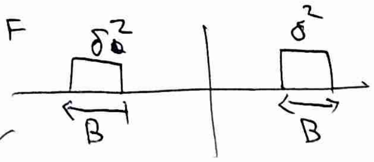
·) $SNR = \dfrac{\text{Power } x(t)}{\text{Power } N(t)}$

→ In real life there is always a filter in our hardware so we never observe the full spectrum of the Noise

$Y(t) = (X(t) + N(t)) * \underset{\text{filter}}{h(t)}$

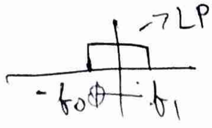
·) We are starting with an ~~uncor~~ ~~uncorrl~~ uncorrelated function
↳ $\delta(\tau)$ but after you filter the Correlation changes.
The Samples are still from gausian distribution but they are correlated

·) LTI System Cannot Convert an uncorrelated signal to correlated

Spectrum of noise after BPF 

↘ Power → integrate with f

Power = $\sigma^2 B$

→ LPF 

$B = f_0 + f_1$

Power = σB

New class

Why do we define ~~δ~~ $R_{NN}(\tau) = \delta(\tau)$ instead of $\begin{cases} 1 & \tau=0 \\ 0 & \tau\neq 0 \end{cases}$

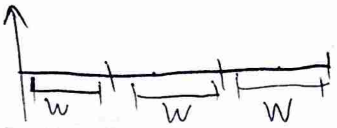
$\delta(\tau)$ ↓ Power is not zero

$\begin{cases} 1 & \tau=0 \\ 0 & \tau\neq 0 \end{cases}$ ↓ you get Power 0

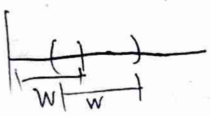
Lim = 0 at $\tau = 0$
But we have 1 at $\tau = 0$

We are looking at Band limited Power spectrum so we want it to be $\delta(\tau)$

• Computing PSD

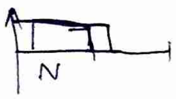


→ Take FFT for W samples and do average (Bartlett)



→ overlapped window (Welsch) then take FFT

~~In~~ IRL

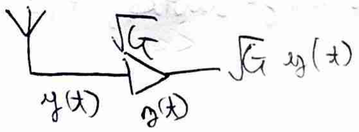


~~Rxx~~ $R_{xx}(m) = \frac{1}{N}\sum_{i=1}^{N} x[i]\,x[i-m]$ → after this do fft.

• Say $R_{NN}(\tau) = N_0\,\delta(\tau)$, $N_0 = KT$ → Noise

Noise Power $= N_0 B$ ↳ Bandwidth (including -ve freq)

• ~~Does~~ An amplifier reduces SNR



$\sqrt{G}(y) = \sqrt{G}\, y(t) + \sqrt{G}\, n(t)$
↓
P_s

$z(t) = \sqrt{G}\, y(t) + N_1(t)$

→ Noise added by amplifier

$\text{SNR output} = \dfrac{G P_s}{G N_0 B + N_1 B}$ → Power adds → Independent.

$R_{YY} = E\left\{ \left(X(t_1) + N(t_1)\right)\left(X^*(t_2) + N^*(t_2)\right) \right\}$

FFT ↓ $= R_{XX} + R_{NN} + R_{XN} + R_{NX}$

$S_Y = S_X + S_N + C_{XN} + C_{NX}$ → Cross ~~pro~~ power spectral density

↳ Can be complex as well $(R_{XY}(\tau) = R^*_{YX}(-\tau))$

$C_{XN} + C_{NX}$ together is real

↓ conjugate Symmetric

Check

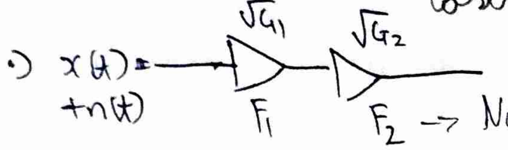
- ~~Noise figure~~ Noise factor $= \dfrac{SNR_{i/p}}{SNR_{o/p}} \geq 1$

- Noise figure (dB) $= NF = SNR_{i/p} - SNR_{o/p}$

  $10 \log_{10}(P)$ — write dBm, dBW

- Open WRT to change transmit power of any router

Cascade ↓

- 

$\sqrt{G_1}\,x(t) + n_1(t) + \sqrt{G_1}\,n(t)$ ↘ $\sqrt{G_1G_2}\,x(t) + n_2(t) + \sqrt{G_1G_2}\,n(t) + \sqrt{G_2}\,n_1(t)$

$SNR = \dfrac{G_1G_2P_s}{G_1G_2N_0B + G_2N_1B + N_2B}$ ↓ output

$$F = \frac{SNR_{i/p}}{SNR_{o/p}} = \frac{P_s/N_0B}{\dfrac{G_1G_2P_s}{G_1G_2N_0B + G_2N_1B + N_2B}} = \frac{1}{N_0G_1G_2B} \times (G_1G_2N_0B + G_2N_1B + N_2B)$$

$$= 1 + \frac{N_1}{N_0G_1} + \frac{N_2}{N_0G_1G_2}$$

- for a ~~gain~~ single device, $F_1 = \dfrac{P_s/N_0B}{\dfrac{GP_s}{GN_0B + N_1B}} = 1 + \dfrac{N_1}{GN_0}$

  $F_2 = 1 + \dfrac{N_2}{G_2N_0}$ → Considering N_0 is given in input

  using this we get $\boxed{F_1 + \dfrac{F_2 - 1}{G_1}}$

$$F_{cascaded} = F_1 + \sum_{i=2}^{n} \frac{F_i - 1}{\prod_{j=1} G_j}$$ Friis formula

- Can we use some LTI block to maximize SNR

  $x(t) \rightarrow \boxed{h(t)} \rightarrow$ ↓ Matched filter

  $SNR = \dfrac{P_x}{P_n} = \dfrac{R_{xx}(0)}{N_0B}$ ↗ $\int_{-\infty}^{\infty} S_x(f)\,df = R_{xx}(0)$

-> A Low noise Amplifier (LNA) in very narrow band

LNA
LPF digital

-> In a digital setting we can Increase SNR

↳ -> LPF ~ -> This happens when the spectrum of noise is present outside the spectrum of the signal -> we can reduce that to improve SNR, But in the same band you cannot remove noise.

-> New class

-> What filter can we use to improve SNR

$$y(t) = x(t) + n(t)$$

$$h(t) * y(t) \rightarrow \text{improves SNR}$$

$$h(t) * y(t) = \underbrace{h(t) * x(t)}_{z(t)} + h(t) * n(t)$$

$$P_N = \sigma^2 \int |H(f)|^2 df \quad \text{for } n \sim N(0, \sigma^2)$$

$$R_{zz^*}(\tau)\Big|_{\tau=0} \rightarrow \int_{-\infty}^{\infty} S_x(f)\,df = R_{xx}(0)$$

↳ Power from PSD

$$R_{zz}(\tau) = \mathbb{E}\left\{\left(x(t+\tau) * h(t+\tau)\right)\left(x^*(t) * h(t)\right)\right\}\Big|_{\tau=0} \rightarrow \text{Substitute}$$

Assuming R_{zz} (depends on τ) then t can be anything (put $t=0$)

$$R_{zz} = \mathbb{E}\left\{\iint \underbrace{x(v) h(-v)}_{g(v)}\, \underbrace{x^*(s) h^*(-s)}_{g^*(s)}\, ds\, dv\right\}$$

↓ Inner product, it is maximized $g(v) = g^*(s)$ -> check

So we have $h(t) = x^*(-t)$ -> we cannot construct this as our message signal is random (If we knew the message whats the point of communication)

-> $x(t) = m(t) \cdot p(t)$

$m(t)$ ↓ Random message

$p(t)$ ↓ carrier, Deterministic

assumption $m(t)$ is constant in the interval where $h(t)$ is matched to $p(t)$

$$= \iint E\{m\,t\}\, p(v) h(-v)\, p^*(s) h^*(-s)\, ds\, dv$$

$h(t) = p^*(-t)$

Also we have already used Matched filter before → demodulation we multiplied the carrier where $p^*(-t)$ is the ~~same~~ carrier only $e^{-j2\pi f_c t}$

- $N(t) = N_I(t) + j N_Q(t)$ → N_I, N_Q (independent, uncorrelated and ~~real~~ jointly WSS for N to be WSS)

$$R_{nn} = \mathbb{E}\left\{ \left(N_I(t_1) + jN_Q(t_1)\right)\left(N_I(t_2) - jN_Q(t_2)\right) \right\}$$

$$= \mathbb{E}\left\{ (N_I(t_1) N_I(t_2) + N_Q(t_1) N_Q(t_2) \right)$$

$R_{nn}(\tau) = R_{N_I}(\tau) + R_{N_Q}(\tau)$

$N_0\,\delta(\tau)$, if the noise is IID, then $R_{N_I} = R_{N_Q} = \frac{N_0}{2}\,\delta(\tau)$

$\sim \mathbb{C}N(0, N_0)$ → complex gaussian

$\sim N(0, \frac{N_0}{2})$ → real Gaussian

- If LO is not synchronized then power in individual arm is different

- SNR in the Passband

→ after demodulation we have $A_c\, m(t) + n(t)$ $\quad SNR = \frac{A_c^2 P_m}{N_0 B}$ → Bandwidth

Noise Power in In Passband

$N_p = N_I(t) \cos 2\pi f_c t - N_Q(t) \sin 2\pi f_c t$ → this is always real

Cyclostationary individually but together they are not

~~If~~ is N_p WSS? $\quad E\{N_p\} = 0$

$$R_{N_p} = \mathbb{E}\{ N_p(t_1)\, N_p(t_2) \}$$

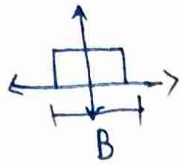
$$R_{N_P} \; \{ N_I(t_1) N_I(t_2)\left[\cos(2\pi f_c(t_1-t_2)) + \cos 2\pi f_c(t_1+t_2)\right]$$
$$+ N_Q(t_1) N_Q(t_2)\left[\cos(2\pi f_c(t_1-t_2)) - \cos 2\pi f_c(t_1+t_2)\right] \} \rightarrow ⓐ$$

So $R_{N_P} = E\{\frac{1}{2} ⓐ\}$, as noise are identical

$= \frac{1}{2}\left[R_{N_I} + R_{N_Q}\right]$ $\quad t_1 - t_2 = \tau$

New class

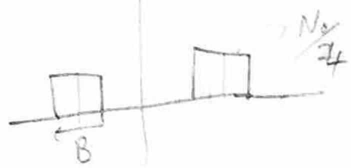
- we can put a LPF or BPF after we receive to get rid of $t_1 + t_2$ terms
- Say $N_I = N_R(t) * h(t) \rightarrow$



$\rightarrow$ Power $= \frac{N_0}{2} B$

$N_R(\tau) = \frac{N_0}{2}\delta(\tau)$

$$R_{N_P} = (R_{N_R}(\tau) * h(\tau) * h^*(-\tau)) \times \cos 2\pi f_c \tau$$
$$\hookrightarrow \frac{\delta(f-f_c) + \delta(f+f_c)}{2}$$



$\rightarrow$ Power $= \frac{N_0}{4} \times 2B = \frac{N_0}{2} B$

The power remains the same? It's just that the components are not orthogonal so its not like their power adds up.

- AM Signals

$A_c m(t) \cos 2\pi f_c t + N_P(t)$

In the base band we have $A_c m(t) + N_I(t) \rightarrow$ in the $\cos 2\pi f_c t$ or I arm

$SNR_P = \frac{A_c^2 P_m/2}{N_0 B/2}$, $SNR_b = \frac{A_c^2 P_m}{N_0 B/2} \rightarrow$ DSB-SC

Convolution with cos add the power up.

DSB-AM

$A_c(1+\alpha m(t))\cos 2\pi f_c t + N_p(t)$

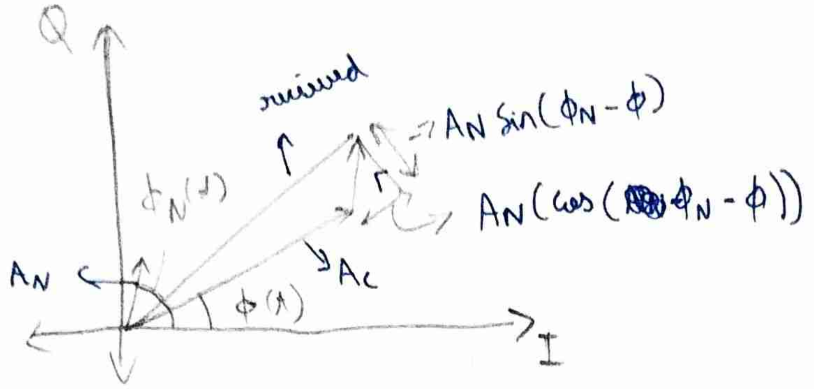
$$SNR_P = \frac{\frac{A_c^2}{2}(1+\alpha^2 P_m)}{\frac{N_0 B}{2}}$$

$$SNR_b = \frac{A_c^2 \alpha^2 P_m}{N_0 B/2} = \frac{2\alpha^2 P_m}{1+\alpha^2 P_m} SNR_P$$

Baseband SNR drops a lot for $\alpha < 1$, this is because a lot of Power is used in the carrier.

FM

$A_c \cos(2\pi f_c t + \phi(t)) + N_p(t)$
↳ convert to $A_N(t)\cos(2\pi f_c t + \phi_n(t))$



$$y(t) = \sqrt{(A_c + A_N\cos(\cdot\cdot))^2 + A_N^2\sin(\cdot\cdot)}\,\cos\left(2\pi f_c t + \phi(t) + \tan^{-1}\left(\frac{A_N\sin(\cdot\cdot)}{(A_c + A_N)\cos(\cdot\cdot)}\right)\right)$$

~~if we assume~~ Our message info is present only in the phase.

If $A_c >> A_N$

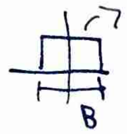
$\cancel{y(t)} \approx \phi(t) + \frac{A_n(t)}{A_c}\sin(\phi_n(t) - \phi(t))$ → SNR increases with A_c

~~assume~~

$\Rightarrow \frac{1}{A_c}\{[A_N \sin\phi_n(t)\cos(\phi) - \cos(\phi_n(t))\sin(\phi)]\}$

assume $\phi_n(t)$ varies a lot more that ϕ → so assume constant

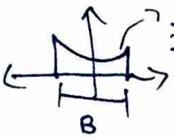
this is ~~called~~ just rotation so



⇒ for FM we take a derivative of phase

$|H(f)|^2 = f^2$

$\frac{1}{2\pi}\frac{d}{dt}$ ↗

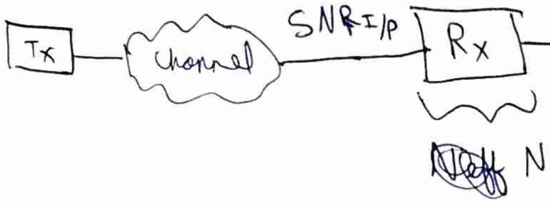


$$\frac{N_0}{2A_c^2}\int_{-B/2}^{B/2} f^2\, df = \frac{B^3}{12}$$

$SNR_{FM} = K$ Photo on Phone

New ~~class~~ class

Link budget analysis



→ $SNR_{o/p}$ (You want this)

- $y(t) = h\,x(t) + n(t)$

  h → ~~att~~ attenuation by channel

- Propagation loss (for free space) $= \left(\frac{4\pi d}{\lambda}\right)^2 = \frac{P_{Tx}}{P_{Rx}}$ → $P_{Rx} = \frac{P_T}{\left(\frac{4\pi d}{\lambda}\right)^2}$ → inverse square law

- We can find NF_{eff} for Channel + R_x

  $SNR_{o/p} = 5dB$ → $SNR_{ip} = NF + SNR_{op}$

  $T_{xp} = 60dB + P_s$ → P_s = 15dBm

  60dB → 1Km away, 4π wavelength.

Digital Communication

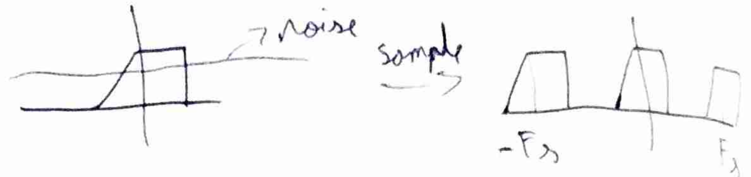
→ allows us to communicate at -ve SNR as you can map bits to bits.

$x(t) \longrightarrow x[n]$

$\downarrow$

B $\qquad F_s > 2B$ (Nyquist)

•) What if there is noise ?



does the noise power go to ∞ ?

•) No because you have to calculate PSD

•) In Practice we add an anti aliasing filter

$x[n] + N[n]$

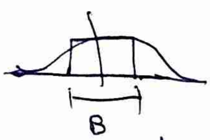
$R_x[\tau] \rightarrow \delta[\tau] \xrightarrow{DFT} PSD$

$\downarrow$ $\qquad\qquad \downarrow$

B $\qquad\qquad$ or DTFT check ?

•) Noise effective Bandwidth

after a filter ↳



there is noise outside B which all aliases when we sample

We just compute

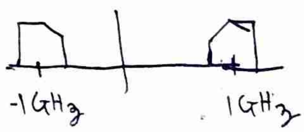
$$B_N = \frac{\int_{-\infty}^{\infty} S_N(f)\,df}{S_{max}(f)} \qquad N_0 = KTB_N$$

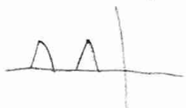
•) What should we sample this at

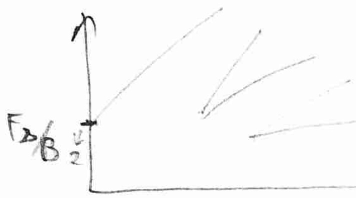
↳ there is a range

↓

Band Pass Sampling

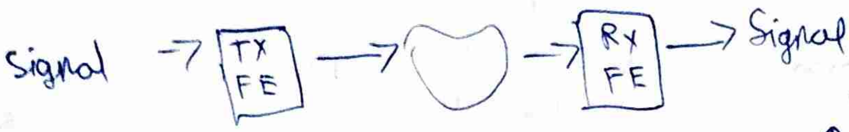




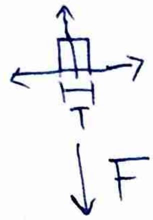


New class

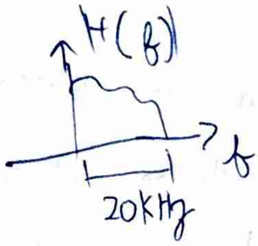
•) About the competition



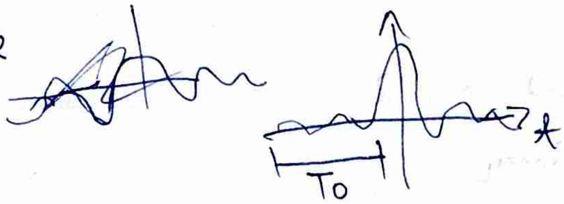
BPSK



we want to use.



we want to pick



Sink ~~spectrum~~ with

•) 99.99% energy, we have a latency of To as we need to know 0 or 1 before transmitting (Non Causal).

•) We estimate the $\hat{H}(f)$ at the reciever, find $\hat{h}^{-1}(t) = f^{-1}(\hat{H}^{-1}(f))$ then you are sending

$$x(t) = \sum_{k=0}^{\infty} b[k] * p(t-nT)$$, we send $x(t) * \hat{h}^{-1}(t)$

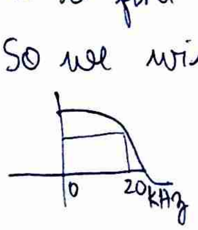
↓ Pulse ↓ Precoding

why at the ~~reciever~~ transmitter? because we convolve with noise at the reciever

↳ do at transmitter because reciever will have noise

•) You can do matched filtering with a pulse shape

•) to find $\hat{H}^{-1}(f)$ you sent a $\delta(t)$ but we canot send this so we will send $\text{Sinc}(t)$ where ~~pulse~~ is in frequency



we get Something above 20KHz

•) But we have currently only ~~20Kb~~ 20 kbps, to increase this, we can use ~~the~~ complex signals. But in real domain only we can have more levels of pulse. 4 levels 00, 01, 10, 11

•) we can only 16 levels or 8 bits max → quantizer of reciever

·) ~~Can~~ you can ML $x(t) \rightarrow \boxed{Tx-Chn-Rx} \rightarrow y(t)$

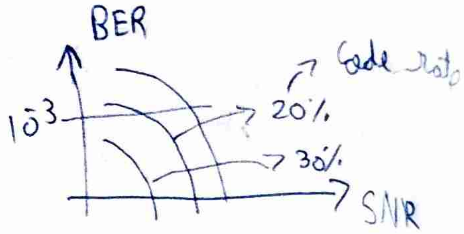
you can use ML

train with regression $y(t) \rightarrow \boxed{\quad} \rightarrow x(t)$

·) Let us try for BER (Bit error rate) $\leq 10^{-3}$

·) Try reed Solomon or hamming Codes

·) How much Error coding do you need



pick a BER and then Based on SNR
pick your ~~reduced~~ redundancy

·) Synchronize the transmitter and reciever → Send a pseudo random sequence where auto correlation is zero ~~on~~ if shift is even a small amount.

Quiz 2

Q1 a) KTB = $\underline{-101\ dBm}$

b) SNR = $\underline{101\ dB}$

c) SNR o/p = 91 dB

d) $L = \left(\frac{4\pi d}{\lambda}\right)^2 \approx 60\,dB$

$24 - 60 = -36 + 10\log \overset{\rightarrow Noise}{N}$

Noise = thermal + interference
thermal → negligible

e) $F_1 + \frac{F_2 - 1}{G_1} \approx F_1$

Q2 Cyclostationary process $R_x(t+T,\tau) = R_x(t,\tau)$
↳ it is two directional

average it over τ to get a periodic (Cyclostationary) process
↳ $\bar{R}(t)$, $\bar{R}(\tau)$ is average over t
↳ it is not WSS

↳ have to check the mean is also periodic

Q3 ~~down~~ doesn't matter if we measure noise in Passband / baseband

$N(t) = N_S \cos 2\pi f_c t - N_Q \sin(2\pi f_c t) = A(t) \cos(2\pi f_c t + \Theta(t))$

$N_I + j N_Q$

variance of $\Theta \sim \text{Unif}[-\pi, \pi]$

$E[\Theta(t+\tau)\,\Theta(t)] = \frac{\pi^2}{3} \delta(\tau)$

mean of A^2

$E\{(N_I^2(t+\tau) + N_Q^2(t+\tau))(N_I^2(t) + N_Q^2(t))\}$ → as they are IId

$= 2E\{N_I^2(t+\tau) N_I^2(t)\} + 2E\{N_I^2(t+\tau) N_Q^2(t)\}$ ↓ discussed below - ⓐ

$=$ ~~2~~ (curtosis something)

→ Band Unlimited

$= 2 \left(\begin{matrix} 3\sigma^4 & \tau = 0 \\ \sigma^4 & \end{matrix} \right.$ → 4th order momentum

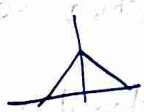
$= 4\sigma^4(\delta(\tau) + 1)$ → PSD is the F.T of this

↳ they are not a zero mean distribution so it is not a $\delta(t)$,

N_I^2 is zero mean but $N_I^2 + N_Q^2$ is not

• The above is for bandunlimited N_I, N_Q,

Band limited ↓

N_I →  → $N_I^2 =$ 

$N_I^2 + N_Q^2$ → the PSD doesn't add UP, the fourier transform of the signal does but not PSD

↳ this doesn't go to zero even if these signals are independant, because they might not have a zero mean → cross terms are not zero

think of this ~~as~~ in terms of Jensens inequality $f(x(t)) = x^2(t)$

$\therefore E[f(N_I), f(N_Q)]$ →

ⓐ → $2E\{N_I^2(t+\tau)\} E\{N_Q^2(t+\tau)\} = 2R_{N_I}^2(0)$

i) Now ~~So~~ how to handle $E\{N_I^2(t+\tau)N_I^2(t)\}$

Isserlis theorem

$$E\{X_1X_2X_3X_4\} = E[X_1X_2]E[X_3X_4] + E[X_1X_3]E[X_2X_4] + E[X_1X_4]E[X_2X_3]$$

after solving $4N_0^2(R^2(\tau)+R_N^2(0))$

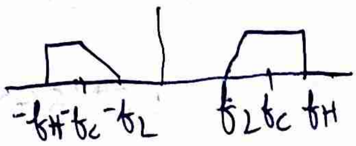
$\rightarrow$ this is the $\triangle$ + additional stuff

Q4 $X(t)$ we get it at $t=nT_s$

$$\underbrace{S[k]}_{1} + \underbrace{N[k]}_{N_0}$$

$$-10\log N_0 \geq 4 \Rightarrow N_0 \leq 3.98$$

i) Bandpass Sampling



We do not want overlap so we have these conditions

$-f_L + kf_s \leq f_L$ the kth component shouldn't overlap

$\rightarrow$ anything before this comes before

$-f_H + (k+1)f_s \geq f_H$ $\rightarrow$ anything after this comes after

if this is true for one component then it is true for all components

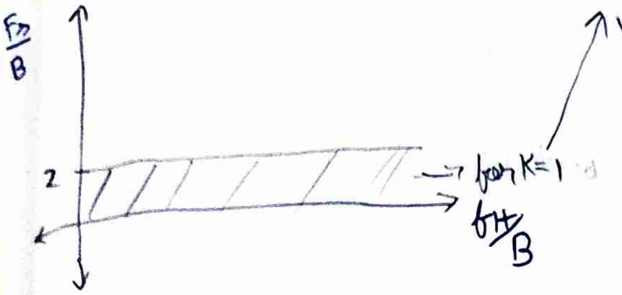
$$\Rightarrow \frac{2f_H}{k+1} \leq f_s \leq \frac{2f_L}{k}$$

$$\Rightarrow \frac{2f_H}{k} \leq f_s \leq \frac{2f_L}{k-1} \rightarrow \text{need to find } k$$

$$\hookrightarrow \frac{2f_H}{k} \leq \frac{2f_L}{k-1} \Rightarrow \frac{1}{f_L} \geq k\left[\frac{1}{f_L}-\frac{1}{f_H}\right]$$

or $k \leq \dfrac{f_H}{f_H - f_L} \Rightarrow \boxed{k \leq \dfrac{f_H}{B}}$

from the inequality $\frac{2f_H}{kB} \le \frac{F_s}{B} \le \frac{2f_L}{(k-1)B}$



$$2 \le \frac{2f_H}{kB} \le \frac{F_s}{B} \le \frac{2f_L}{(k-1)B} = \frac{2(f_H/B - 1)}{k-1}$$

if k=0, when f_H is comparable to B → overlap occurs

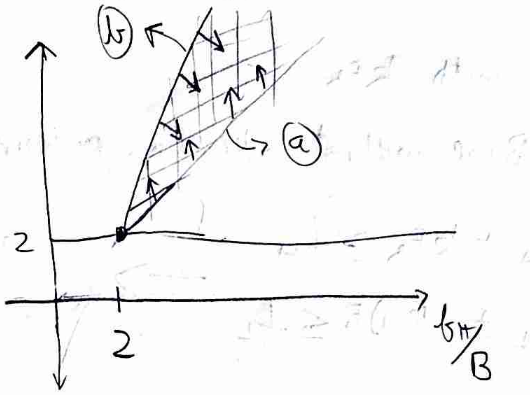
if k=1, $\frac{F_s}{B} \le \infty$

now for every k, $2\left(\frac{f_H}{B} - 1\right)$ is the upper bound

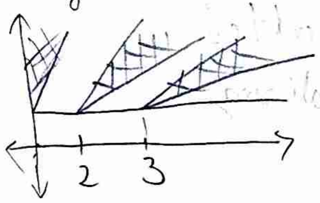
$2\left(\frac{f_H}{B}\right)$ is the lower bound.

if $f_H >> B \Rightarrow f_H \approx f_L$ or the ~~gab~~ gap between allowed regions for sampling reduces, small shift in LO causes aliasing

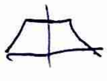
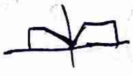
for k=2 $\underbrace{\frac{2f_H}{2B}}_{a} \le \frac{F_s}{B} \le \frac{2f_L}{B} = \underbrace{2\left(\frac{f_H}{B} - 1\right)}_{b}$



For any k

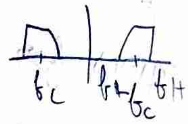


ask lekha

if we get  instead of  when sampling how do we filp it ? ↗

Newclass

- we need 2 parameters f_c and Bondwidth (B) or f_L and f_H

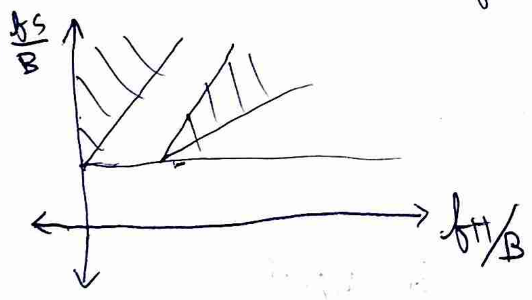


- Sample with kF_s
- $F_s \geq B$ we is the bare minimum

$$-f_H + kF_s \geq f_H$$
$$-f_L + (k-1)F_s \leq f_L \implies \frac{2(f_c + B/2)}{k} \leq f_s \leq \frac{2(f_c - B/2)}{k-1}$$

and $f_s \geq B$

- We make a plot of 2 variables by normalizing

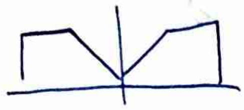


- How to find k $\quad \frac{2f_H}{k} \leq \frac{2(f_H - B)}{k-1}$ f_L

$\Rightarrow k \leq \frac{f_H}{B}$

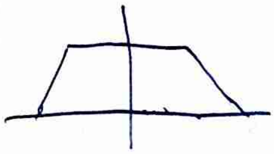
as at 2 4B ↳ ask manoj why

When you sample you get one of the two below, how do you convert one to another?

•)  $\rightarrow$ $-f_L + nf_s = 0$ $\quad nf_s = f_L \le \frac{2n f_L}{k-1}$

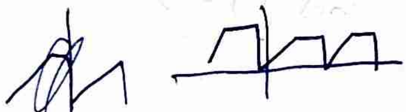
$k-1 \le 2n$

$\rightarrow$ K odd $\rightarrow$ how check

•)  $-f_H + nf_s = 0$ $\quad \rightarrow k$ even

$f_H = nf_s \ge \frac{2n f_H}{k} \rightarrow k \ge 2n$

•) Say you have

 how to get  $\rightarrow$ shift by $B/2$

$X(f)$ $\qquad\qquad X(f - B/2)$

Similarly
in time domain multiply $e^{j2\pi t f_0} = e^{j 2\pi n T_s B}$ $\quad \rightarrow f_0 = B$

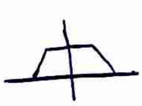
$= (e^{j\pi})^n$ $\quad$ assuming $F_s = 2B$, $T_s = \frac{1}{F_s}$

$\downarrow$

$+1$ or -1 in time domain

 $\rightarrow$ Shift by B to get 

↳ Check

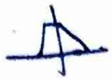
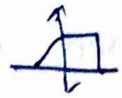
•) for  $\rightarrow e^{j 2\pi \frac{n}{F_s} \times \frac{F_s}{4}} = (e^{j\frac{\pi}{2}})^n$

$\downarrow$

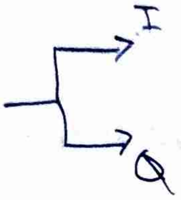
$1, j, -1, -j \dots$

•) here you do not need to multiply by $\cos(2\pi f_c t)$ as we can just ~~S~~ multiply by complex exponential.

•) For noise $\rightarrow$ We ~~fist~~ first pass through BPF

•) if you want  but you recieve 

this is because of I, Q arm getting ~~g~~ different, the delay is different?



→ I/Q ~~[illegible]~~ imbalance, $X_I + jcX_Q$

↓ Imaginary axis is flipped

•) After we do Sampling there is Quantization

$$\hat{x}[n] = Q\{x[n]\} \rightarrow \text{non-linear}$$

•) Quantization causes loss of information

$\hat{x} = x + w$ → this is the quantization noise

$w = -\tilde{x} + x$

$$SQNR = \frac{P_x}{P_w}$$ Can this be ∞? means $P_w = 0$

•) What if our input signal itself is discrete? then w can be zero? humm

•) We want to ~~[illegible]~~ analytically evaluate SQNR

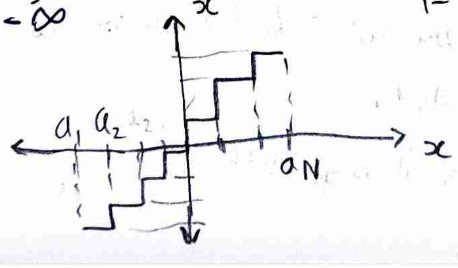
•) Assumptions → zero mean and uncorrelated

if it was correlated you will have to find PSD

•) $\mathbb{E}\{\underbrace{(x-\tilde{x})^2}_{\text{Distortion}}\}$

•) $D = \int_{-\infty}^{a_1} (x-\tilde{x}_1)^2 P(x)\,dx + \sum_{i=1}^{N-1} \int_{a_i}^{a_{i+1}} (x-\hat{x}_{i+1})^2 P(x)\,dx + \int_{a_{N-1}}^{\infty} (x-\hat{x}_N) P(x)\,dx$

distortion measure



- $N = 2^b$

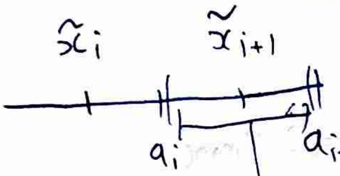
we can change $a_1 \dots a_{N-1}, \tilde{x}_1 \dots \tilde{x}_N$ to optimize our quantization ↳ it is discrete in time, we make it discrete in amplitude also

- $\tilde{x}_i$ can be 0000 or 1111 or how do you pick these

$$\frac{\partial D}{\partial a_i} = -(a_i - \tilde{x}_{i+1})^2 P(a_i) + (a_i - \tilde{x}_i)^2 P(a_i) = 0$$

$$\Rightarrow \boxed{a_i = \frac{\tilde{x}_{i+1} + \tilde{x}_i}{2}} \quad \text{— (a)}$$

→ independent of probability distribution



↳ the bounds are equidistant

- now $$\frac{\partial D}{\partial \tilde{x}_i} = -\int_{a_{i-1}}^{a_i} 2(x - \tilde{x}_i) P(x) = 0$$

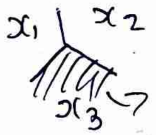
$$= \boxed{\tilde{x}_i = \frac{\int_{a_{i-1}}^{a_i} x P(x)\, dx}{\int_{a_{i-1}}^{a_i} P(x)\, dx}} \quad \text{— (b)}$$

→ Conditional expectation

- ~~Day~~ Lloyd-Max optimizer → you need to alternate between (a) and (b) a_i and $\tilde{x}_i$ and keep optimizing untill you converge

this is scalar optimization

- You can also do ~~vector~~ optimization for higher dimentions, set of all ~~direct~~ points closest to any point



→ voronoi for x_3 → Used in computer graphics

↳ needed to quantize complex numbers

• ~~...~~ There is Uniform and non uniform quantization

let us fix $q_{i+1} - q_i = \Delta$

this implies $\rightarrow x_i$ is fixed (it is equidistant from q_{i+1} and q_i)

$$D = \int_{-\infty}^{q_1} (x - q_1 - \Delta/2)^2 P(x)\,dx + \sum_{i=1}^{N-1} \int_{q_1+(i-1)\Delta}^{q_1+i\Delta} (x - (q_1 + i\Delta - \Delta/2))^2 P(x)\,dx$$

$$+ \int_{q_{N-1}}^{\infty} (x - (q_{N-1} + \Delta/2))^2 P(x)\,dx$$

• Let us fix $x \in [-x_{max}, x_{max}]$ → Dynamic range $= 2x_{max}$

$q_0 = -x_{max}$, $q_N = +x_{max}$

now $\Delta = \frac{2x_{max}}{N}$

if we assume $P(x)$ is Unif, $P(x) = \frac{1}{2x_{max}} = \frac{1}{\Delta N}$

Substituting this we get $D = N \int_{-\Delta/2}^{\Delta/2} y^2 \times \frac{1}{\Delta N}\,dy = \frac{\Delta^2}{12}$

How did we get y^2

New ~~class~~ class

now $SQNR = \frac{\Delta^2}{12} = \frac{3N^2}{\left(\frac{x_{max}^2}{P_x}\right)} \approx \frac{3 \cdot 4^b}{\alpha}$ → PARP

SQNR in dB $= 4.8 + 6b - 10\log\alpha$ α is always > 1

• 1 bit increase gives us 6 dB extra SQNR

now if $b = 0$ that means we have a constant as output, this is because we are still calculating error $(C - x_{max})^2$

- Our ~~au~~ audio signals are gaussian distribution → our assumption that x is uniform is wrong.
- we need to transform our signal to look like uniform to keep our ADC ~~is~~ optimal

$x \longrightarrow f(x)$ → ~~Via~~ You need to transform

$x \downarrow N(\cdot)$ $\quad f(x) \hookrightarrow u_{\text{rand}}$

$\boxed{\text{Companding}}$ $\qquad$ $\boxed{\text{A- law}}$, Box - Muller Transform

$$f(x) = \text{Sign}(x) \times \frac{1+\log(A|x|)}{1+\log(A)} \rightarrow \text{if } |x| \geq \frac{1}{A}$$

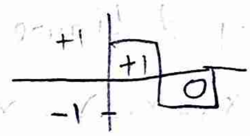
$$\text{Sign}(x) \times \frac{A|x|}{1+\log A} \rightarrow \text{else}$$

→ you multiply by Something → low numbers are higher vice versa

overall gaussian $\times f(x)$ looks approx uniform

<u>PCM</u> (Pulse Code Modulation)

$x(t) \rightarrow x[n] \rightarrow Q(x[n]) \rightarrow \{b_i\}$



- In old CD, DVD ~~1D~~ 1 hour of Audio ~~@~~ used to ~~be~~ be 600 MB

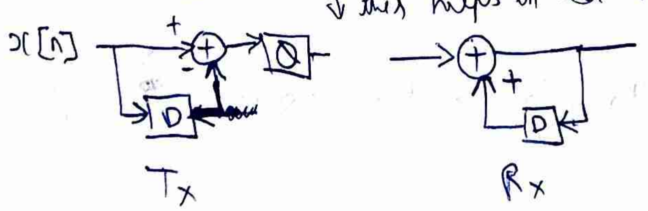
$40 \times 10^3 \times 3600 \times 8$ bit = 144 MB

$F_s = 40\text{KHz}$; 1 Hour

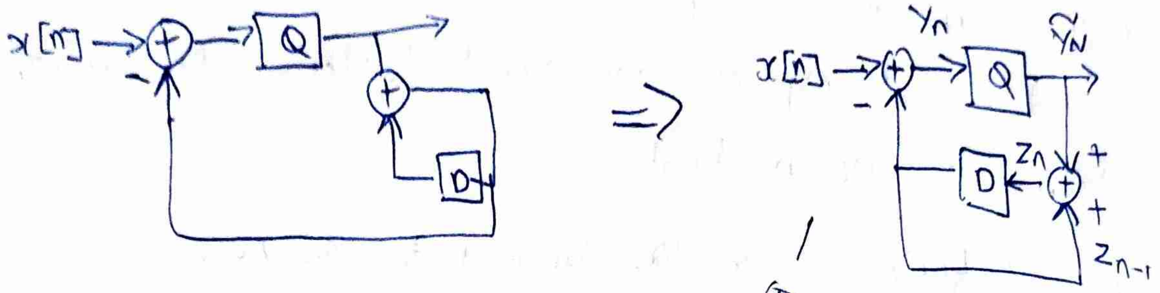
- MP3 file used to store audio in freq domain instead of time → Check how? How to store
- ML can be used to find non linear transform to compress other that F.T

- How to change ↗ reduce Dynamic range (Change α to ↑SQNR)

<u>Differential PCM</u> → We made an assumption that the samples are uncorrelated → Wrong
↓ this helps in ~~that~~ that $x[n] - x[n-1]$



- We get a better performance in, we are doing $x_n - \tilde{x}_{n-1}$



(a)

new class

Idea for DPCM → $\max[x_n - x_{n-1}] < \max x_n$ — (b)

will this increase PARP?

Say $E\{x_n^2\} = P_x$ $\quad E\{(\underset{\text{error}}{x_n - x_{n-1}})^2\} = 2P_x - 2E\{x_n x_{n-1}\}$

$= 2P_x(1-\rho)$

we want ρ to be big, so ~~power~~ error ↓ samples are close together

- Compressed sensing → to samples at lower than nyquist rate

- looking at figure (a)

$Q(x_n - x_{n-1}) = x_n - x_{n-1} + w_n$ → this keeps accumulating so we want to reduce this

we do $Q(x_n - \tilde{x}_{n-1})$ → it is reduced

- at the reciever we have

- Compare $\tilde{y}_n - y_n$ and $\hat{x}_n - x_n$

$\tilde{y}_n - y_n = \tilde{y}_n - x_n + Z_{n-1}$

$= Z_n - x_n$

$= \hat{x}_n - x_n$ $\quad \Rightarrow Z_n = \hat{x}_n$ as we have just copy pasted the reciever in (a)

This is true if $Z_0 = X_0 = 0$

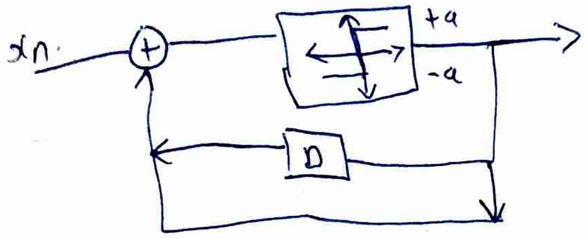
$\therefore \tilde{y}_n - y_n = \hat{x}_n - x_n$

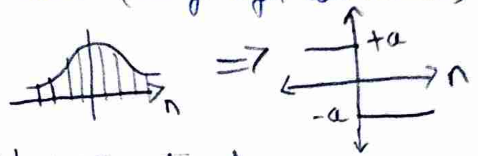
- But now SQNR of $(\tilde{y}_n - y_n)$ is > than SQNR $(\hat{x}_n - x_n)$

as from (b)

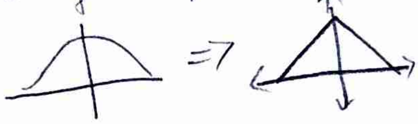
Delta Modulator

- we can do 1 bit ~~two~~ quantization by oversampling → ~~our~~ our signal looks like a straight line approx (very high correlated)





at high Sample rate



- There is an Slope overload, if signal has slope greater than $\frac{a}{T_s}$ then problem, so increase T_s

Sample at T_s, max height $= a \times \frac{T}{T_s}$

Max slope $= \frac{a \times \frac{T}{T_s}}{T} = \frac{a}{T_s}$

- $Y_n = X_n + W_n$, $\underset{\downarrow \text{ estimate of } X_n}{\hat{X}_n = g(Y_n)}$ ⇒ $Pr(\hat{X}_n = X_n) = 0$ in analog com

$\nearrow P(W_n = 0) = 0$

- what if $x_n \in \{-1, 1\}$ and $g(Y_n) = \text{Sign}(y)$

$$Pr(\hat{X}_n = X_n) = Pr(\overset{\alpha}{X_n = +1})\, Pr(W_n > -1) + \underset{1-\alpha}{Pr(X_n = -1)}\, \underset{\text{symmetric}}{Pr(W_n < 1)}$$

$$= Pr(W_n > -1)$$

↳ greater than zero

- Signal Space analysis

$S_1(t) \ldots S_N(t)$

⇓

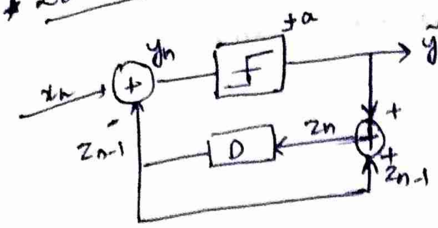
$\Psi_1(t) \ldots \Psi_N(t)$ (orthogonal signals)

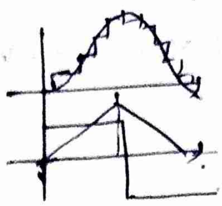
then $\underbrace{\sum a_i \Psi_i(t)}_{x(t)}$

message information

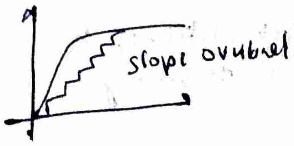
Pulse or Carrier

★ Delta model → 1 bit quantizer → comparator





here $b=1$. & try to reduce α as much as possible
by over sampling.



In a given time T, the largest
slope you can give

$\Delta \times \frac{T}{T_s}$ → max slope = $\frac{\Delta}{T_s}$
$\overline{\quad T \quad}$

but if signal has more slope than
you lower T_s → over sample.

★ **Digital communication**

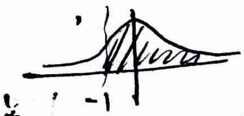
$x(t)$ x_n
$y_n = x_n + w_n$ $\quad \hat{x}_n = g(y_n)$ → estimate of x_n.

~~Pr(...)~~ $Pr(\hat{x}_n = x_n) = Pr(w_n = 0) = 0$
↳ Never get back exact.

$\hat{x}_n = g(y_n)$
$x_n \in \{\pm 1\}$
$\hat{x}_n = g(y_n)$ $\quad g(y) = \text{sign}(y)$

$Pr(\hat{x}_n = x_n) = Pr(x_n = +1)\, Pr(w_n > -1) + Pr(x = -1)\, Pr(w_n < +1)$ — equal due to symmetry of gaussian
($Pr(x_n = +1) \to \alpha$, $Pr(x=-1) \to 1-\alpha$)

$= Pr(w_n > -1)$ ↳ $> \frac{1}{2}$



↓ Can go close to 1 in digital.

↓ $1 - 10^{-3} \approx 0.999$ (~~Practical~~ Practical level of recovery)

★ Signal space:

Which signals can you use for digital comm.

$s_1(t) \ldots s_N(t)$
↓
Construct orthogonal signals.
$\varphi_1(t) \ldots \varphi_N(t)$

Any signal which we transmit will be like $\underbrace{\sum a_i \varphi_i(t)}_{x(t)}$ → data resides here.

★ Digital Modulation

Put the bits on a waveform and recover back the bit sequence.

$b_0 b_1 \dots$ $\quad 0 \to +1$, $1 \to -1$ or anyway.

This cannot be transmitted through wireless

↳ has frequency components all the way from $0 \to$ something large.

cannot get antennas for such low frequency.

Waveform mapping

↳ bits to symbol mapping

↳ pulse shaping.

$b_0 b_1 \dots \longrightarrow x(t)$

what bit sequence, what $x(t)$ to use depends on medium.

↳ $s_1(t)\ s_2(t) \dots s_M(t) \longrightarrow$ bunch of waveforms depending on your wave-form you choose.

$b s_1(t)$

$$x(t) = \sum_i \sum_j a_i s_j(t)$$

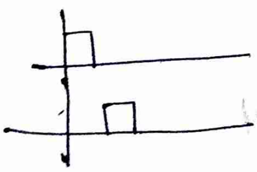
given, you can construct a set of orthogonal waveforms. → easy to seperate signals.

$\psi_1(t)\ \psi_2(t) \dots \psi_M(t)$ (Gram-Schmidt orthogonalization)

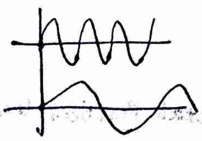
$$\text{Innerproduct} = \int_{-\infty}^{\infty} s_1(t)\, s_2^*(t)\, dt$$

Can choose sinusoids also.

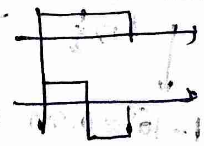
$\psi_1(t) = rect(t/T)$



Time division multiplexing



Freq. div. mux.



Code div. mux.

$$x(t) = b_1\psi_1(t) + b_2\psi_2(t) + \dots$$

$$= \sum_i b_i \psi_i(t)$$

b_i → carries information.

$av_1 + bv_2 = x.$

to get a & b

→ take inner product with v_1 & v_2 sep. ~~basis~~

$\frac{\langle x, v_i \rangle}{\|v_i\|^2} = a_i$

because of this we take orthonormal waveforms
↓
divide by energy.

Just because ∞'ly many orthogonal waveforms exist in a given bandwidth doesn't mean you can send ∞ bits.

$\sum a_i v_i + \underset{\text{noise}}{n} = x.$

$a_i + \langle v_i, n \rangle \langle x_i, n \rangle$

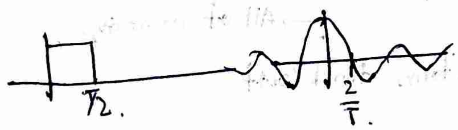
Why not? why no ∞ data rate?

→ In practice, you cannot do $-\infty$ to ∞.
so you put a limit from $0 \to T$, and with this period you cannot get ∞ many signals.

when only sinusoid from $-\infty$ to ∞

from $0 \to T$.
eq to multiplying with rectangle pulse.
so very small bandwidth not possible.

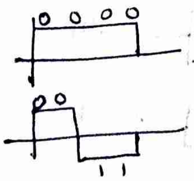
For



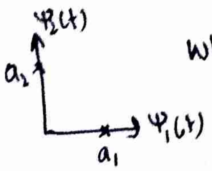
$f = \frac{k}{T}$

If ∞ orthogonal → ∞ bandwidth
finite → finite

Walsh code:
- Hadamard.
Used in CDMA.



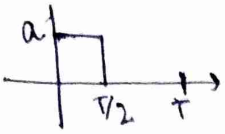
$b_1 b_2 \ldots b_M$
↓
T/M



What ever you transmit will be a point in this vector space.

$x(t) = \sum_{i=1}^{M} b_i \psi_i(t)$ (M degrees of freedom)
↳ ∝ bandwidth ∝ datarate.

$\int_0^T x(t)\psi_i^*(t)\,dt = b_i$



$a^2\, T/2$

So $\frac{\varphi_1(t)}{\sqrt{a^2 T/2}}$ → time duration is very small then scaling factor ~~becomes~~ becomes very large.

ALL ARE ENERGY SIGNALS HERE → NO POWER SIGNALS

CDMA → has a Rake receiver

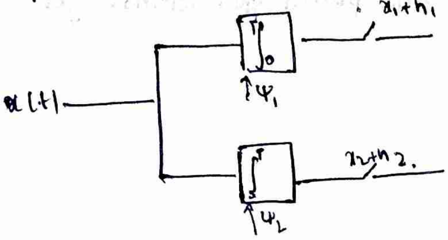
different arms.

$\int_0^T x(t)\varphi_i^*(t)\,dt = b_i + n_i$ → Match filtering process
↓
Process that enables you to maximize SNR.

$n_i = \int_0^T n(t)\varphi_i^*(t)\,dt$ → Random process through LTI but output is gaussian but not white.
AWGN → independent

↓ But when you sample it, it might become white.

Impulse response of a match filter of $h_i(t) = \varphi_i^*(T-t)$



eventually you end up working with samples that have only white noise

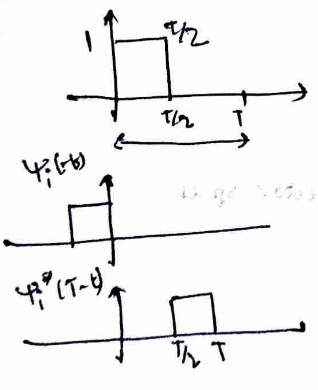
$t=T \int_0^T x(\tau) * h(T-\tau)\,d\tau = \int_0^T x(t)\varphi_i^*(t)\,dt \quad h(t)=?$

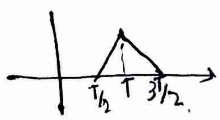
sin cos.

If you have 2 orthonormal waveforms → QAM

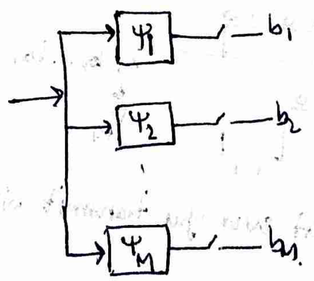
Signal space modelling → orthonormal waveforms find coeff
→ All of comms.
↓
Research → how do you pick the right optimal space to get better BW eff, etc.

Okay, now for the matched filter $\varphi_i^*(T-t)$, $x(t) * \varphi_i^*(T-t)$





Sample at T, exactly there you'll get the signal you're looking for



How do you know you are looking at $[0,T]$ and sampling at $T_s = T$.

For any $x(t)$ $[0,T]$ can be anywhere, Local oscillators not synchronized.

All these things work, only if everything is synchronized.

$\langle x(t) \mid \varphi_j(t) \rangle$

gen at Tx — gen at R_x

These two in general not synchronized.

SNR = $\dfrac{\text{Signal En.}}{\text{Noise En.}} = \dfrac{\int_0^T |x(t) * h(t)|^2 \, dt}{N_0 \cdot \int |h(t)|^2 \, dt}$ → conv. evaluated only at $t = T$. → $\mathbb{E}\{|z(T)|^2\}$ → you sample it a T bcz finally you work with discrete samples only.

where $z(t) = x(t) * y(t)$

signal here is $x(t) * h(t)$

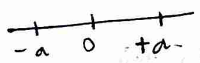
Assume $h(t) = c\,\varphi(T-t)$ → The SNR will be independent of c. So scaling doesnot matter.

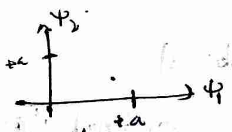
$= \dfrac{E_s}{N_0}$ → E_s → Energy of coeff $b_j[n]$; N_0 → PSD

Say you want to transmit: 0 1 1 0 1 1 1 1

M = 1

$\varphi_1(t)$





Performance will be acc. to additive noise.

Noise it can move ⟷ only along 1D

Noise will be in 2D, much larger error space

How well can you distinguish 2 clusters of error

2a sep — better

$\sqrt{2}\,a$ sep

Analytically, → sampled at T

$b + n_1$, $n_1 \sim (0, \sigma^2)$

$b = \pm a$ — $N_0 \times$ filter area

$b \sim \begin{cases} +a & \text{with prob } p \\ -a & \text{with prob } 1-p \end{cases}$

$P(b=+a \mid w) = \dfrac{P(w \mid b=+a)\, P(b=+a)}{P(w)}$ $P(b=-a \mid w) = \dfrac{P(w \mid b=-a)\, P(b=-a)}{P(w)}$

$P(b=+a|w) \overset{?}{>} P(b=-a|w)$

$P(w|b=+a)\,P(b=+a) \overset{?}{>} P(w|b=-a)\,P(b=-a)$

$$w|b \sim \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(w-b)^2}{2\sigma^2}\right)$$

$$\Rightarrow \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(w-a)^2}{2\sigma^2}\right) p > \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(w+a)^2}{2\sigma^2}\right)(1-p)$$

$$\Rightarrow \exp\left(\frac{-(w-a)^2+(w+a)^2}{2\sigma^2}\right) > \frac{1-p}{p}$$

$$\Rightarrow \frac{-(w-a)^2+(w+a)^2}{2\sigma^2} > \log_e\left(\frac{1-p}{p}\right)$$

$$\Rightarrow \frac{4a.w}{2\sigma^2} > \ln\left(\frac{1-p}{p}\right) \Rightarrow w > \frac{\beta\sigma^2}{2a} \Big\} \beta$$

$\underbrace{\ln\left(\frac{1-p}{p}\right)}_{\beta}$ $\qquad$ $>0.$

$$P(b=+a|w) = \frac{P(w|b=+a)\,P(b=+a)}{P(w|b=+a)\,P(b=+a) + P(w|b=-a)\,P(b=-a)}$$

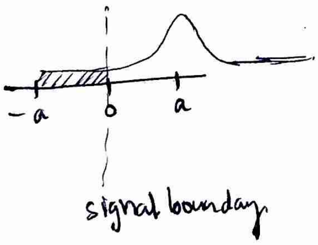
$$= \text{sigmoid}(\alpha) = \frac{1}{1+e^{-\alpha}}$$

→ optimal classifier for this thing, cannot construct a better one.

log likelihood ratio.

$$\alpha = \ln\left(\frac{P(w|b=+a)\,P(b=+a)}{P(w|b=-a)\,P(b=-a)}\right)$$

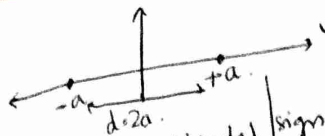
→ likelihood.



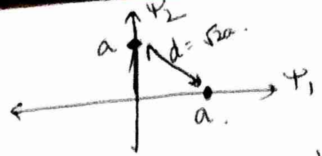
→ defn. for standard normal fn.

→ Q function??

$P(n>p) = Q($

★ 

Binary antipodal signaling.
→ sigmoid classifier.



Binary orthogonal signaling.

$y = x + n$, $n \to \mathcal{N}(0, \sigma^2)$

$x = \begin{cases} +a & \text{w p } p \\ -a & 1-p \end{cases}$

$P(y|x) = \mathcal{N}(a, \sigma^2)$

$y > \alpha$

$P_{error} = \cancel{Re} \; P_r(W > -a) \, Pr(x = +a) + Pr(W > a) \, Pr(x = -a)$

$\approx Q\left(\sqrt{\frac{E_s}{\sigma^2}}\right)$ $\quad E_s = a^2 \longrightarrow Q\left(\frac{d}{2\sigma}\right)$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ a \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

MAP decision if you know $P_r(x = +a)$, $Pr(x = -a)$

$P(x = s_1 | y) > P(x = s_2 | y)$

$y_1 > y_2$

$P_e = Pr(\hat{x} = s_1 | x = s_2)$

$\to Pr(n_2 - n_1 > a)$

$Pr(y_2 - y_1 > 0)$ sum of two gaussians is always gaussian given they are uncorrelated.

$-a + n_2 - n_1 \sim \mathcal{N}(-a,$

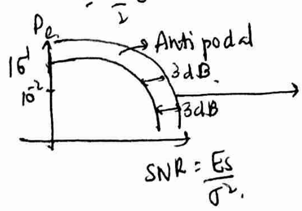
$y_1 = a + n_1$
$y_2 = 0 + n_2$ $|_{s_1}$

$E_s = a^2 = \frac{d^2}{2}$

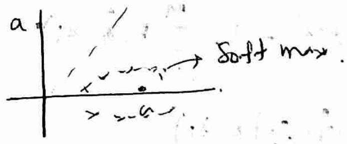
$P_e = Pr(\hat{x} = S_1 | x = S_2) \to Q\left(\sqrt{\frac{d^2}{4\sigma^2}}\right) = Q\left(\frac{d}{2\sigma}\right) \longrightarrow$ Farther the distance, the better it is.

$= Q\left(\sqrt{\frac{E_s}{2\sigma^2}}\right)$ (you should send more power here than $Q\left(\sqrt{\frac{E_s}{\sigma^2}}\right)$ case to get the same data rate)

$= \frac{1}{2} e^{-E_s/4\sigma^2}$



~~Circled~~ Circularly symmetric gaussian noise

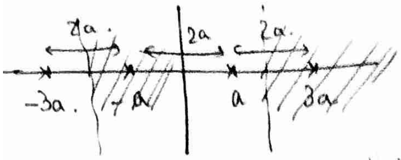


Linearly seperable data → communication system.

~~$\|y - S_i\|^2$~~ $\arg\min_i \|y - S_i\|^2$ = demod. → this is the optimal thing.
↳ 1-nearest neighbour.

→ C different constellation points.

$P_{er} = \sum_{i=1}^{C} P_r(\hat{x} \neq s_i | x = s_i) \, P(x = s_i) = \sum_{i=1}^{C} (1 - P(\hat{x} = s_i | x = s_i)) \, P(x = s_i)$



↳ decision boundary

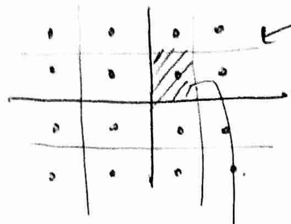
To find what signal, find out posterior probability

↓

Whichever has maximum that is best

Pulse Amplitude Modulation (PAM)

PAM in orthogonal basis



4 in each of the axis

↳ QAM.

M-PAM ← wired communi.

M-QAM ← wireless.

$$Pr\left(\begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \in R\right)$$

★ $Pr(\hat{x} \neq s_i | x = s_i) = Pr\left(\bigcup_{s_i \neq s_j} \hat{x} = s_j \,|\, x = s_i\right)$

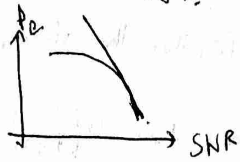
$\leq \sum_{s_j \neq s_i} Pr(\hat{x} = s_j | x = s_i)$ ← Mutually exclusive events but still why do we put inequality

$\leq \sum Q\left(\frac{d_{ij}}{2\sigma}\right)$ ↳ If you use optimal demodulator then equality works ore else inequality.

$\sum_{s_j \neq s_i} Q\left(\frac{d_{min}}{2\sigma}\right)$

$d_{min} = \min_{i,j} d_{ij}$

$(M-1) Q\left(\frac{d_{min}}{2\sigma}\right) \rightarrow$ upper bound is tight.



★ $Pr(\hat{x} = s_1 | x = s_2) = Q\left(\frac{d}{2\sigma}\right)$ M constellation points

$P_e \leq P(\hat{x} \neq x) = 1 - \sum_{i=1}^{M} P(\hat{x} = x_i)$

$= \sum_{j \neq 1} \underbrace{Pr(\hat{x} = s_j | x = s_i)}_{Q\left(\frac{d_{ij}}{2\sigma}\right)}$

$P_e = \sum_{i=1}^{M} \sum_{j \neq 1}^{M-1} Pr(\hat{x} = s_j | x = s_i) Pr(x = s_i)$

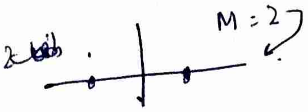
→ Send as many bits as possible
→ Reliably decode the bits sent.

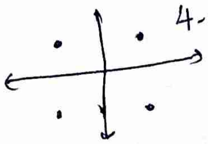
M-PAM & M-QAM

With a single constellation point, you send $\log_2 M$ bits.

$\log_2 M$ ⎫
1, 2, 4, 6, 8
$\underbrace{1,2,4}$ → More commonly used.
6, 8 → very rare.

$M = 2$ → smallest constellation.

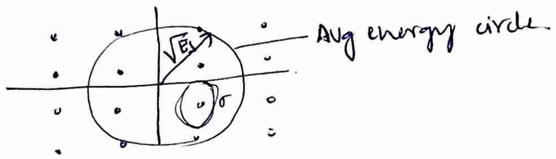




16 & 64 QAM are commonly used.

If you fix d_{min} & inc. constellation points → implies you are increasing energy of your signal.

If the average energy = transmitted energy & more the no. of points you can fit in increase, so transmission rate ↑. optimal → d_{min} can't go less than a certain number.



If you fix SNR ⇒ E_s & σ are getting fixed.

SNR ↓ Max no. of points you can pack is given by

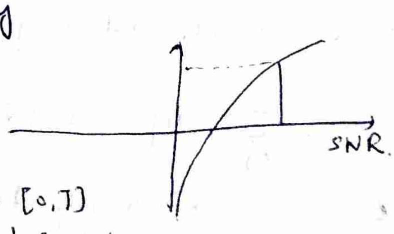
σ ↓ Amount of noise as you go away from a constellation point

$M \approx \log_2(1+SNR)$ → Capacity of AWGN.

If everything is properly source coded, then you get uniform distribution.

How much information you can send = Entropy

Entropy of R.V = $\mathbb{E}(-\log_2 P(x))$ bits.



We usually pick uniform distribution.

$\varphi_1(t)\ \varphi_2(t)$ → Basis from $[0,T]$ → one symbol per $[0,T]$, so in sec, $\frac{1}{T}$ symbols

↳ $\frac{1}{T}\log_2 M = B\log_2 M$ → bit rate.

$B\log_2(1+SNR)$ → bits per sec.

$Pr(\hat{b}_i \neq b_i)$

$b_1\ b_2\ b_3\ b_4 \ldots$

$\underbrace{b_1\ b_2}_{\downarrow s_1}\ \underbrace{b_3\ b_4}_{s_2} \ldots$

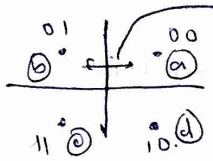
signals

How to get bit error rate?

If $P_{SER} \approx 10^{-3} \longrightarrow$ symbol error rate.

$\hookrightarrow \frac{1}{1000}$.

$P_{BER} = \frac{1 \text{ or } 2??}{2000} \longrightarrow$ depends on how we do bit to symbol mapping



$\longrightarrow$ bit error would be 1 bit

if ⓑ was 11 then 2 bits

$P_{BER} < P_{SER}$ always almost.

but we want to minimize BER, so we always ensure that neighbouring constellation points differ by 1 bit only.

$\downarrow$

grey code

In some cases like PSK

$\downarrow$

orthogonal slightly better than antipodal

In satelliteduodin MFSK

Peak to average power is always the same.

4-PSK = Q-PSK

Interference issues $\longrightarrow$ do frequency multiplexing

Given a const. diagram,

$\downarrow$

coeff of basis fn. for a particular transmitted sign

★ Different types of modulation:

$\{x_1, x_2, \ldots x_M\}$ $\quad \sum_{n=-\infty}^{\infty} \sum_{j=1}^{L} [a_{J_n i}]\, \psi_i(t-nT)$

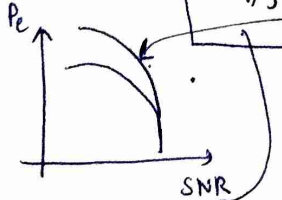
$$\begin{array}{c} \psi_1 \\ \psi_2 \\ \psi_L \end{array} \begin{bmatrix} a_{11} \\ a_{12} \\ a_{1L} \end{bmatrix} \ldots \ldots \begin{bmatrix} a_{M1} \\ \vdots \\ a_{ML} \end{bmatrix}$$

$x_{J_n} \in \mathcal{X}$

$y = x + n$

$Q\left(\frac{d}{2\sigma}\right) \longrightarrow$ probability that x_2 gets confused with x_1 or vice versa

Total error $= \sum_{i \neq j} Q\left(\frac{d_{ij}}{2\sigma}\right) \approx (M-1)\, Q\left(\frac{d_{min}}{2\sigma}\right)$



$\longrightarrow$ Probability of error always indicates symbol error rate. (SER)

Bit error rate (BER) $\approx \frac{SER}{\log_2 M}$ (whenever grey code used)

Reason: here we consider intersection = 0.

$SNR = \frac{E}{\sigma^2}$ → radius within which all points should lie.

Capacity of channel: No. of symbols a channel can accommodate.

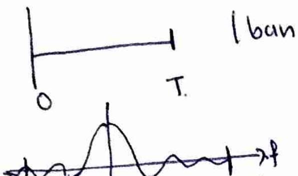
log(1+SNR) → Max. no. of points you can pack in a circle.

↳ No. of bits per sec per Hz bps/Hz.

↓ Spectral efficiency.

Data rate, bit rate = $B\log_2(1+SNR)$ bps.

What happens to the actual bandwidth:



(bandwidth becomes ∞).

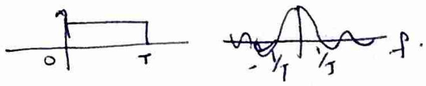
Things die out after a particular frequency.

so does not really matter for low data rates & fibers → because they have a limit of which they transmit.

A good copper cable → 2Gbps.

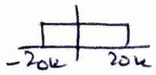
* Pulse shaping: P(t)

Right now $P(t) = rect(t/T)$

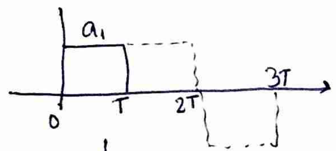


can we do something better?

Our project: Audio :



What is "better"?



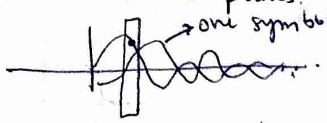
↓

do match filter & sample & recover signal.

Find out pulses that can be used in bandwidth limited signals & still recover back all coeffs.

Nyquist criteria for ISI-free pulses.

Inter symbol interference



If you sample here, you will get sum of all symbol amplitudes.

$z[0] = \sum x[m] P(0-mT) = x[0]$

$z[n] = \sum x[m] P(nT-mT) = x[n]$

If pulse satisfies this then this issue is avoided.

You are going to sample at exactly one point, so if there exists even one point where there is no interference then we can sample it at that point.

$z(t) = \sum_{m=-\infty}^{\infty} x[m] P(t-mT)$

$z[n] = z(nT)$ $\quad P(nT) = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$

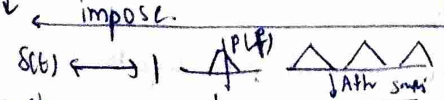
$\boxed{\text{if } P(f) \leq B}$ then sinc pulse is the only pulse that satisfies.

Consequence of this pulse shape:

$P(t) \cdot \sum \delta(t-nT)$

$\leftrightarrow P(f) * \left[\frac{1}{T}\sum \delta(f-\frac{n}{T})\right]$

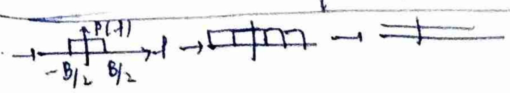
impose.



→ Changes

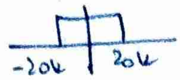
↓

best one is a sinc in time.



Sinc pulse ⟶ best bandwidth efficiency. $\boxed{\text{NYQUIST}}$

Here, sinc is a kind of match filter

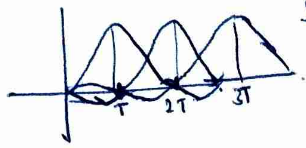
Now ~~ye~~ 

⟶ your channel looks something like this.

So. you have to come up with the best pulse

that has max. SNR ⟶ match filter ~~which~~ ⟶ sinc in time.

Now, given this sampling rate, is there a shape with which I can get a larger bandwidth.



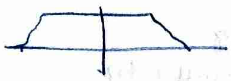
Truncate $(-\infty$ to $\infty)$.

Say 99% of sinc's power is in $[-LT, LT]$

So delay ~~[illegible]~~ your symbol by LT.

↓

Non causality is dealt by delaying.

Practically, channel looks like 

can I do something to get a little more bandwidth.

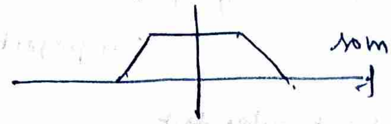
↓

Also localizes the pulse

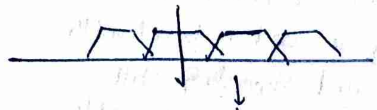
⟶ delay can be reduced.

- Get something that decays faster than sinc.

In frequency domain



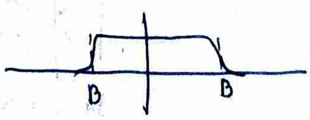
something like this.



↓

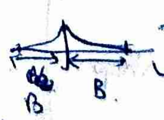
when you add this

Something else is a $\boxed{\text{raised cosine pulse}}$. used in practice.



$$\text{sinc}(t/T)\,\frac{\cos \pi\alpha t/T}{1-4\alpha^2 t^2/T^2}$$

α ⟶ excess bandwidth ratio / roll off bandwidth

largest value of α 

Time domain

min roll off ⟶ same bandwidth

max roll off ⟶ double bandwidth.

So we actually transmit: $\sum_{n=-\infty}^{\infty} x[n]\, \underbrace{p(t-nT)\,\psi(t-nT)}_{\tilde{p}(t-nT)}$

$x[n] \downarrow x_t$

~~[illegible]~~ combine. so that pulse shape does not [illegible] split your pulse so recovery at reciever is simpler.

At the reciever we want $\sum_{n=-\infty}^{\infty} x[n]\, p(t-nT)$

$$p_i(t)\,\psi(t) = \tilde{p}_i(t)$$

Match filter corresponds to $\tilde{p}_i(t)$

but you will get p convolved with itself

to avoid this you split

$$p(t) = p_s(t) \cdot p(t)$$

root raised cosine.

so that after match filtering you have

In which system which pulse shape you are using $\longrightarrow$ Should be very careful.

Pulse shape depends on basis func.

because while projecting it'll be an issue

$$\int x[n]\,\underbrace{\psi(t)\; p(t)\,\psi^*(t)}\,dt$$

Projection operation.

you want to retain orthogonality.

So you have to worry :)